NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

# Clinical Evaluation of Pulmonary Function Testing 2026

Jason Golbin DO, MBA, MS

November 2026

Do.
Make.
Heal.
Innovate.
Reinvent the Future.

1. What is a PFT?
2. How is it performed?
3. What are the different patterns of pulmonary function?
4. How does one interpret a PFT?
5. What is bronchodilator response?
6. What is a methacholine challenge?
7. What is obstructive lung disease, and which diseases are associated?
8. What is restrictive lung disease, and which diseases are associated?
9. Which diseases affect the DLCO?
10. What are the different lung volumes and lung capacities?
11. What are the different types of obstruction, and how to dx?

Source: Course Syllabus

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

Much appreciation to:

**PHIL MARCUS, MD**

# Pulmonary Function Tests

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- Categorize the nature and severity of disease
- Follow the progression of disease
- Measure response to therapy
- Define disability
- Pre-op evaluation
- Screening at risk populations
- Monitoring drug toxicity

# Pulmonary Function Tests

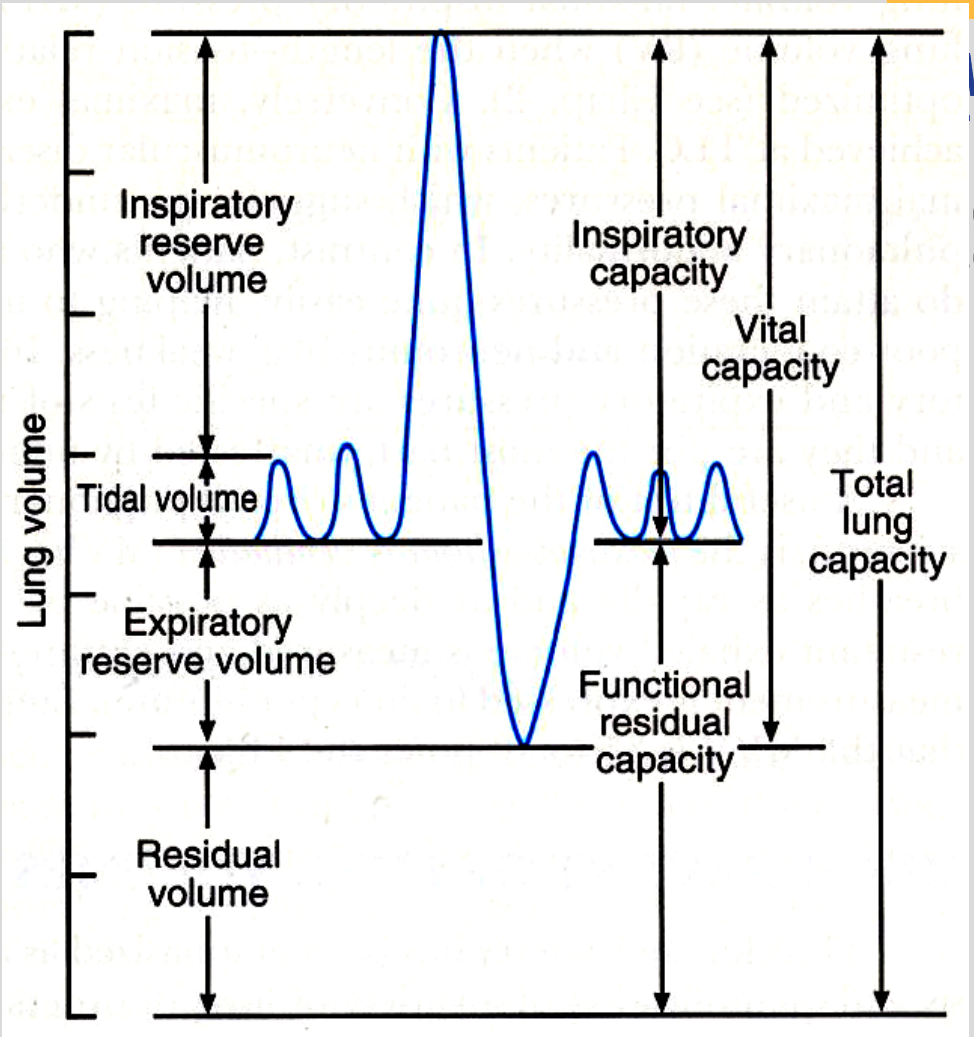
NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- **Tests of Flow**
- **Tests of Volume**
- **Tests of Diffusion Capacity**
- **Tests of Oxygenation**
- **Tests of Ventilation**

# Lung Volumes

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

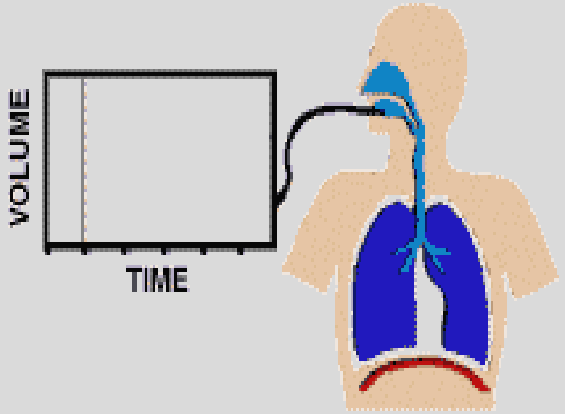
- IRV= Inspiratory Reserve volume
- $V_T$= Tidal Volume
- TLC= Total Lung Capacity
- RV= Residual Volume
- ERV= Expiratory Reserve Volume
- IC = Inspiratory Capacity


Lung volume
Inspiratory reserve volume
Tidal volume
Expiratory reserve volume
Residual volume
Inspiratory capacity
Vital capacity
Functional residual capacity
Total lung capacity

# Spirometry

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- Simple, office-based
- Measures flow, volumes
- Volume vs. Time
- Can determine:
  - Forced expiratory volume in one second ($FEV_1$)
  - Forced vital capacity (FVC)
  - $FEV_1$/FVC



# What Does Spirometry Measure?

- Spirometry measures the rate at which the lung changes volumes during forced breathing maneuvers
- Spirometry begins with a full inhalation and is followed by a forced exhalation that rapidly empties the lung
- Exhalation is continued for as long as possible or until a plateau is reached

Barreiro and Perillo. *Am Fam Physician.* 2004;69:1107-1114.

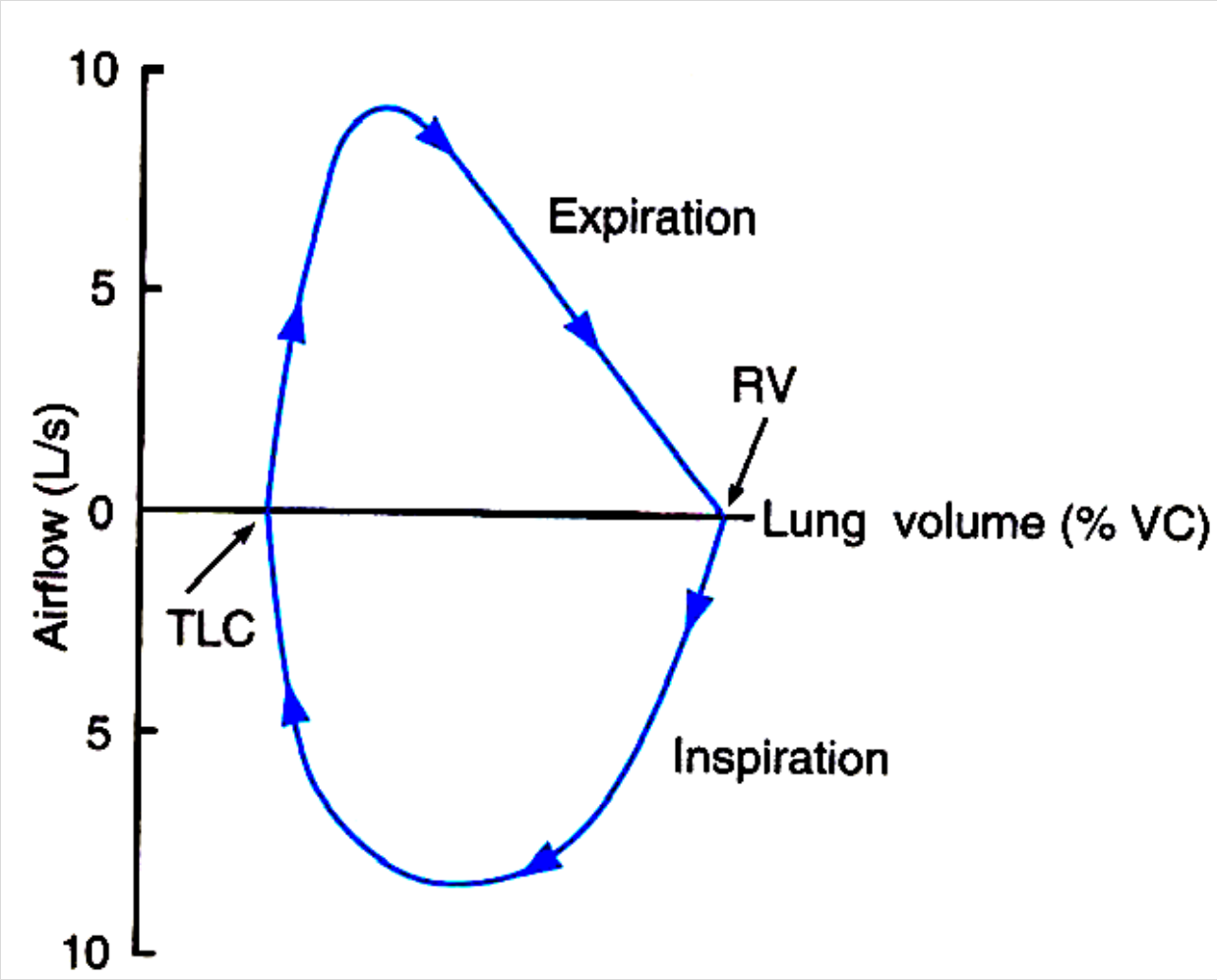
# Interpreting Spirometry Results

**2 ways to graph spirometry maneuvers:**

1. **Volume-time curve**

2. **Flow-volume loop**

Petty and Enright. *Simple Office Spirometry for Primary Care Practitioners.* National Lung Health Education Program. 2003. Available at: http://www.nlhep.org/resources.html#phys. Accessed November 12, 2004.


10
5
0
5
10
Airflow (L/s)
Expiration
RV
Lung volume (% VC)
TLC
Inspiration

# Spirometry

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- FVC = Forced Vital Capacity
- $FEV_1$ = Forced Expiratory Volume in one second
- $FEF_{25-75}$ =
  Forced Expiratory Flow between 75% and 25% of Vital Capacity

# Predicted Values

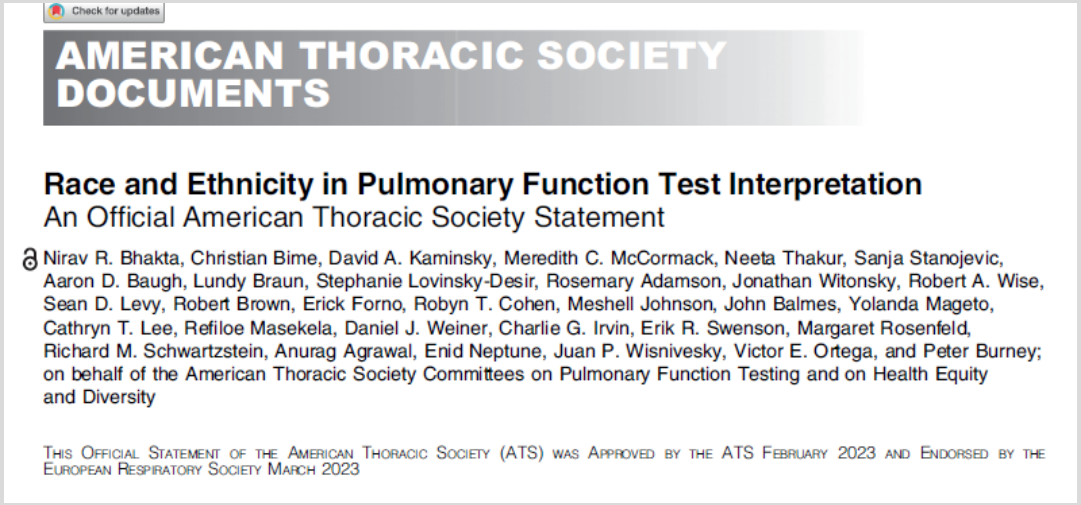
- Age
- Height
- Race
- Gender
- Weight

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

# 2023 Update

College of Osteopathic Medicine



Check for updates

AMERICAN THORACIC SOCIETY DOCUMENTS

**Race and Ethnicity in Pulmonary Function Test Interpretation**
An Official American Thoracic Society Statement

Nirav R. Bhakta, Christian Bime, David A. Kaminsky, Meredith C. McCormack, Neeta Thakur, Sanja Stanojevic, Aaron D. Baugh, Lundy Braun, Stephanie Lovinsky-Desir, Rosemary Adamson, Jonathan Witonsky, Robert A. Wise, Sean D. Levy, Robert Brown, Erick Forno, Robyn T. Cohen, Meshell Johnson, John Balmes, Yolanda Mageto, Cathryn T. Lee, Refiloe Masekela, Daniel J. Weiner, Charlie G. Irvin, Erik R. Swenson, Margaret Rosenfeld, Richard M. Schwartzstein, Anurag Agrawal, Enid Neptune, Juan P. Wisnivesky, Victor E. Ortega, and Peter Burney; on behalf of the American Thoracic Society Committees on Pulmonary Function Testing and on Health Equity and Diversity

This Official Statement of the American Thoracic Society (ATS) was Approved by the ATS February 2023 and Endorsed by the European Respiratory Society March 2023

## Key Conclusions and Recommendations

- PFT laboratories should adopt a race-neutral approach to PFT interpretation by reporting and interpreting results using average reference equations.

American Journal of Respiratory and Critical Care Medicine Volume 207 Number 8 | April 15 2023

# Terminology

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- Forced vital capacity (FVC):
  - Total volume of air that can be exhaled forcefully from TLC
  - The majority of FVC can be exhaled in <3 seconds in normal people, but often is much more prolonged in obstructive diseases
  - Measured in liters (L)

# Forced Vital Capacity

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- Interpretation of % predicted:
  - 80-120% Normal
  - 50-79% Mild reduction
  - 30%-49% Moderate reduction
  - <30% Severe reduction

# Terminology

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- Forced expiratory volume in 1 second: ($FEV_1$)
  - Volume of air forcefully expired from full inflation (TLC) in the first second
  - Measured in liters (L)
  - Normal people can exhale more than 75-80% of their FVC in the first second; thus the FEV1/FVC can be utilized to characterize lung disease

# $FEV_1$ ---(GOLD Criteria)

College of Osteopathic Medicine

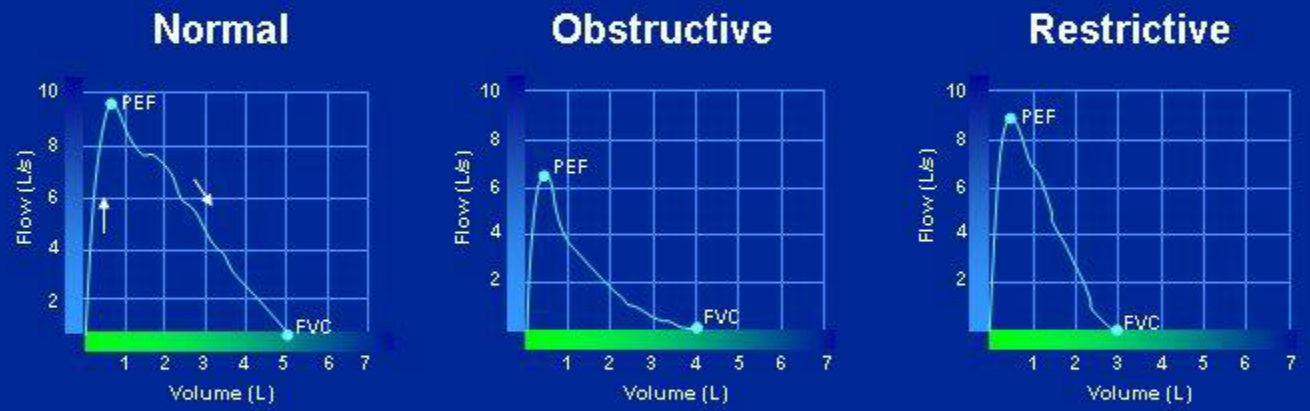
- In presence of a low $FEV_1$/FVC ratio:
- Interpretation of % predicted:
  - >=80% Normal
  - >=50 - <80% Moderate obstruction
  - >=30 - <50% Severe obstruction
  - <30% Very Severe obstruction

# Guiding Principles to Interpreting Pulmonary Function Tests

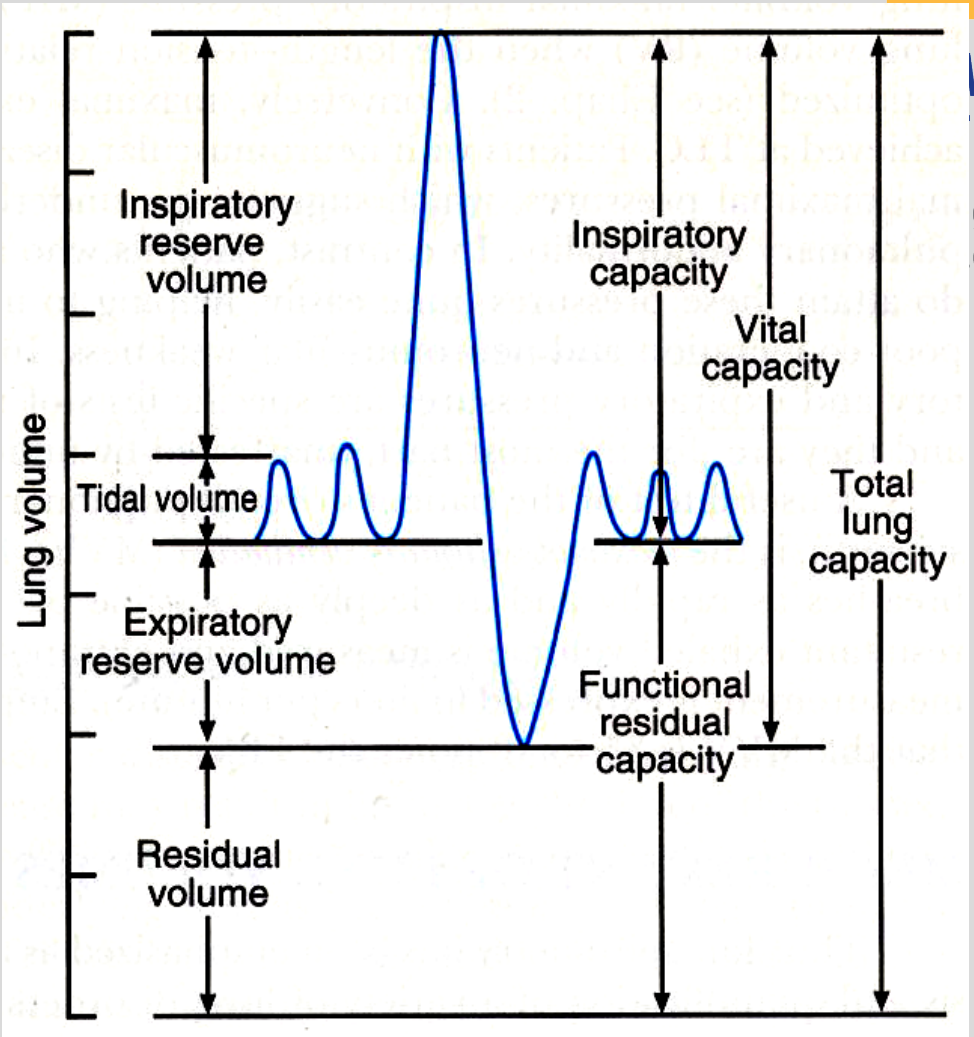
1. Examine the curves
2. Examine the $FEV_1/FVC$ ratio
3. Examine the $FEV_1$ value

Hyatt et al, eds. *Interpretation of Pulmonary Function Tests: A Practical Guide*. Philadelphia, Pa: Lippincott-Raven; 1997.

# Interpreting Spirometry Results: Flow-Volume Curves



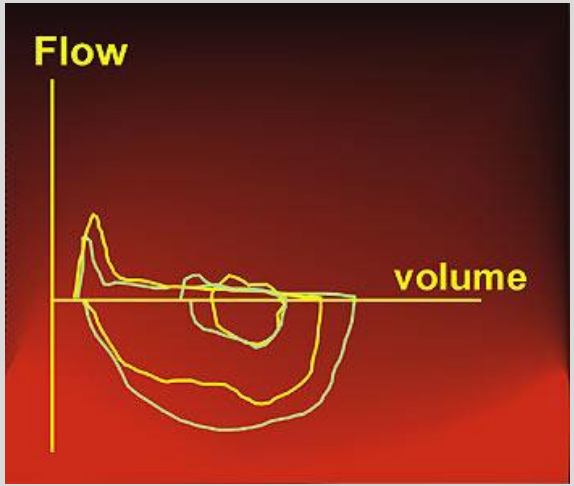

Petty and Enright. *Simple Office Spirometry for Primary Care Practitioners.* National Lung Health Education Program. 2003. Available at: http://www.nlhep.org/resources.html#phys. Accessed November 12, 2004.


Lung volume
Inspiratory reserve volume
Tidal volume
Expiratory reserve volume
Residual volume
Inspiratory capacity
Vital capacity
Functional residual capacity
Total lung capacity

# Obstructive Pattern

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- Decreased $FEV_1$
- Decreased FVC
- Decreased $FEV_1$/FVC
  - <70% predicted
- $FEV_1$ used to follow severity in COPD



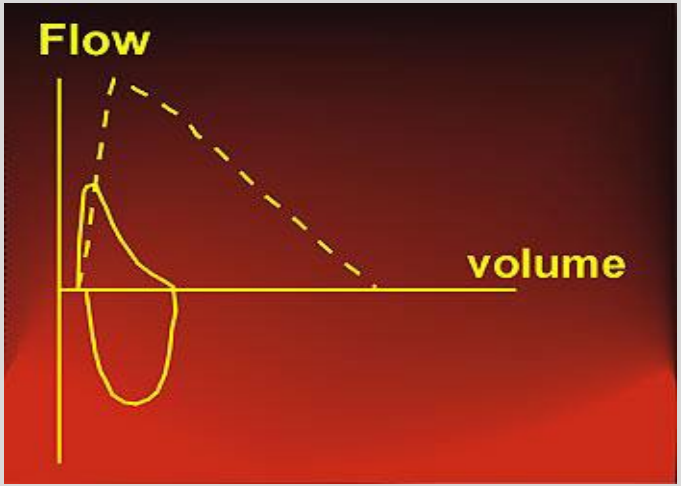
# Obstructive Lung Disease — *Differential Diagnosis*

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

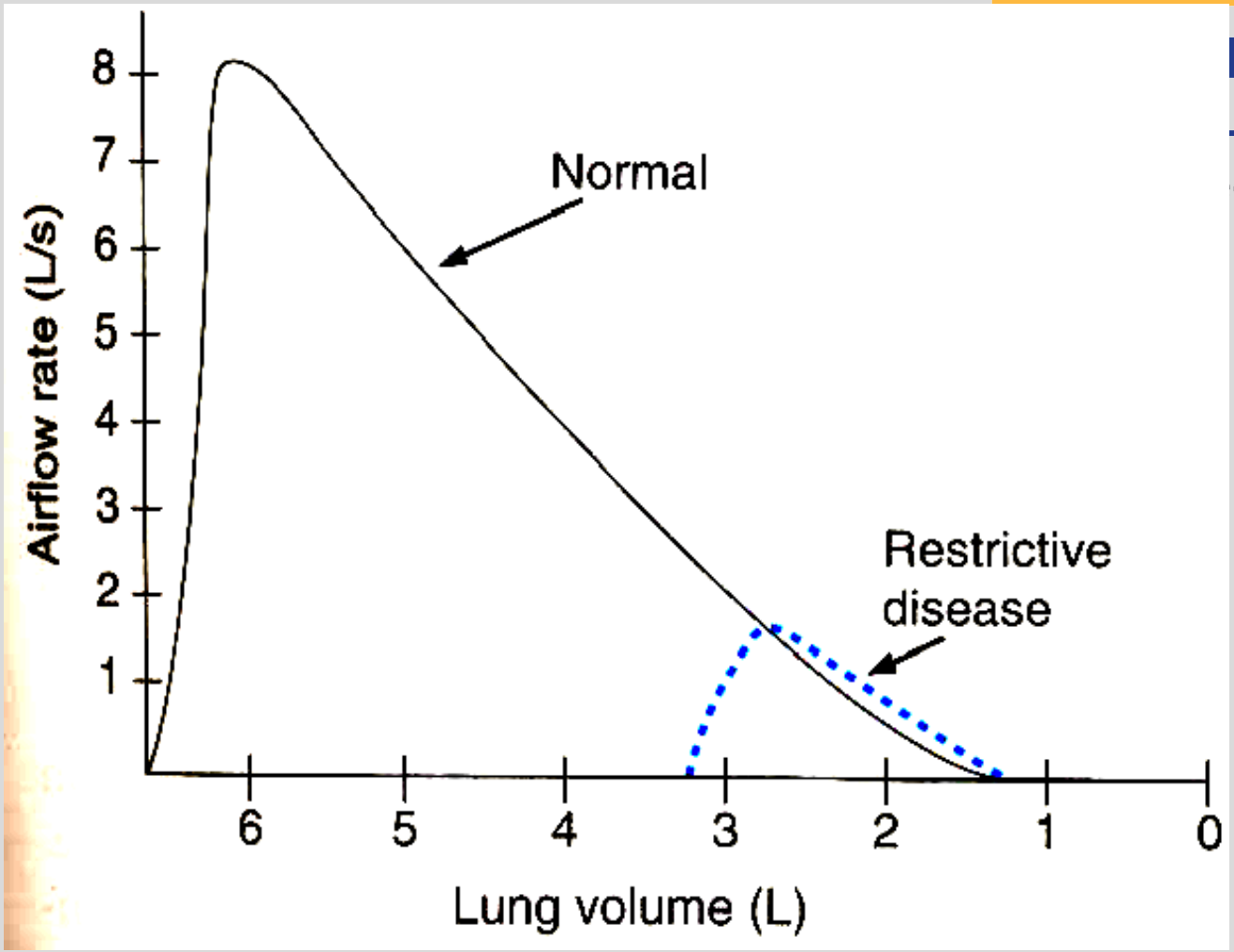
- Asthma
- COPD
  - chronic bronchitis
  - emphysema
- Bronchiectasis
- Bronchiolitis
- Upper airway obstruction

# Restrictive Pattern

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine




- Decreased $FEV_1$
- Decreased FVC
- $FEV_1$/FVC *normal* or *increased*


Airflow rate (L/s)
8
7
6
5
4
3
2
1
Normal
Restrictive
disease
6
5
4
3
2
1
0
Lung volume (L)

# Restrictive Lung Disease

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

***Interstitial Lung Disease***

- Interstitial pneumonitis
- Pneumoconiosis
- Pulmonary Edema

***Space Occupying Lesions***

- Tumors
- Cysts

***Pleural Diseases***

- Pneumothorax
- Hemothorax
- Pleural Effusion
- Fibrothorax

***Chest Wall Diseases***

- Injury/Flail Chest
- Kyphoscoliosis
- Spondylitis
- Neuromuscular Disease

# Restrictive Lung Disease

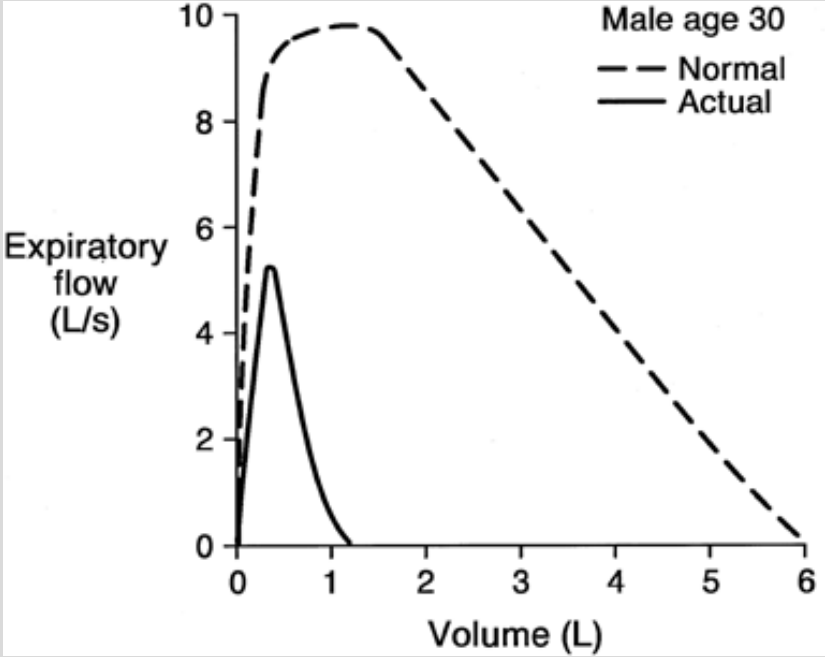
NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

## *Extra Thoracic Conditions*

- Obesity
- Pregnancy
- Ascites
- Peritonitis

# Severe Restrictive Lung Disease

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

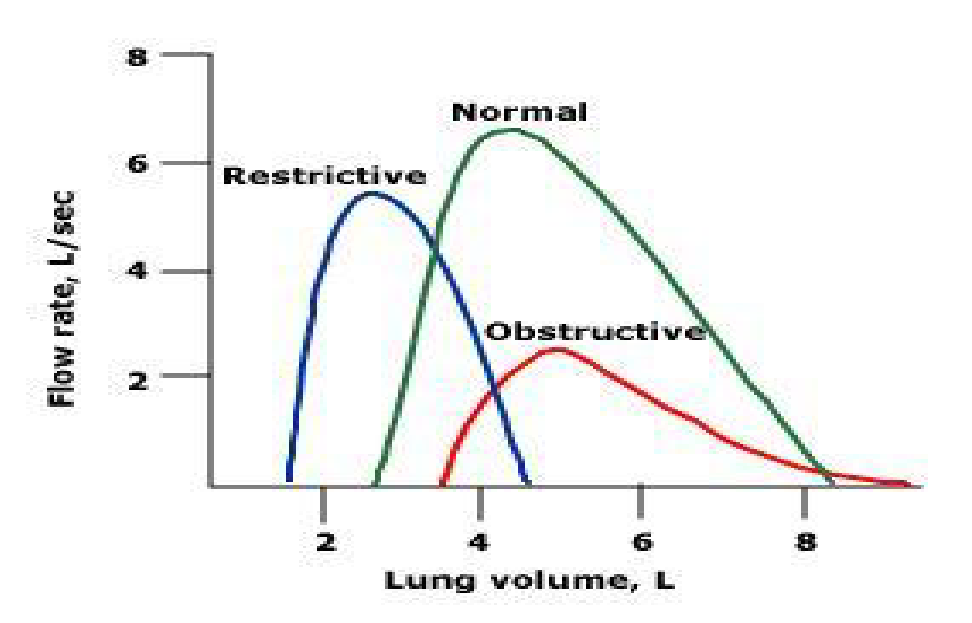



| | | |
|---|---|---|
| TLC | 2.1 L | (28%) |
| FVC | 1.12 L | (19%) |
| $FEV_1$ | 1.04 L | (21%) |
| $FEV_1/FVC$ | 93% | (100%) |
| MVV | 81 L/min | (43%) |
| $D_{L_{CO}}$ | 15 mL/min/mm Hg | (43%) |
| $SaO_2$ | Rest<br>Exercise | 95%<br>88% |
| Slope FV curve | 7.5 | (2-3.0) |

# Spirometry Patterns

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

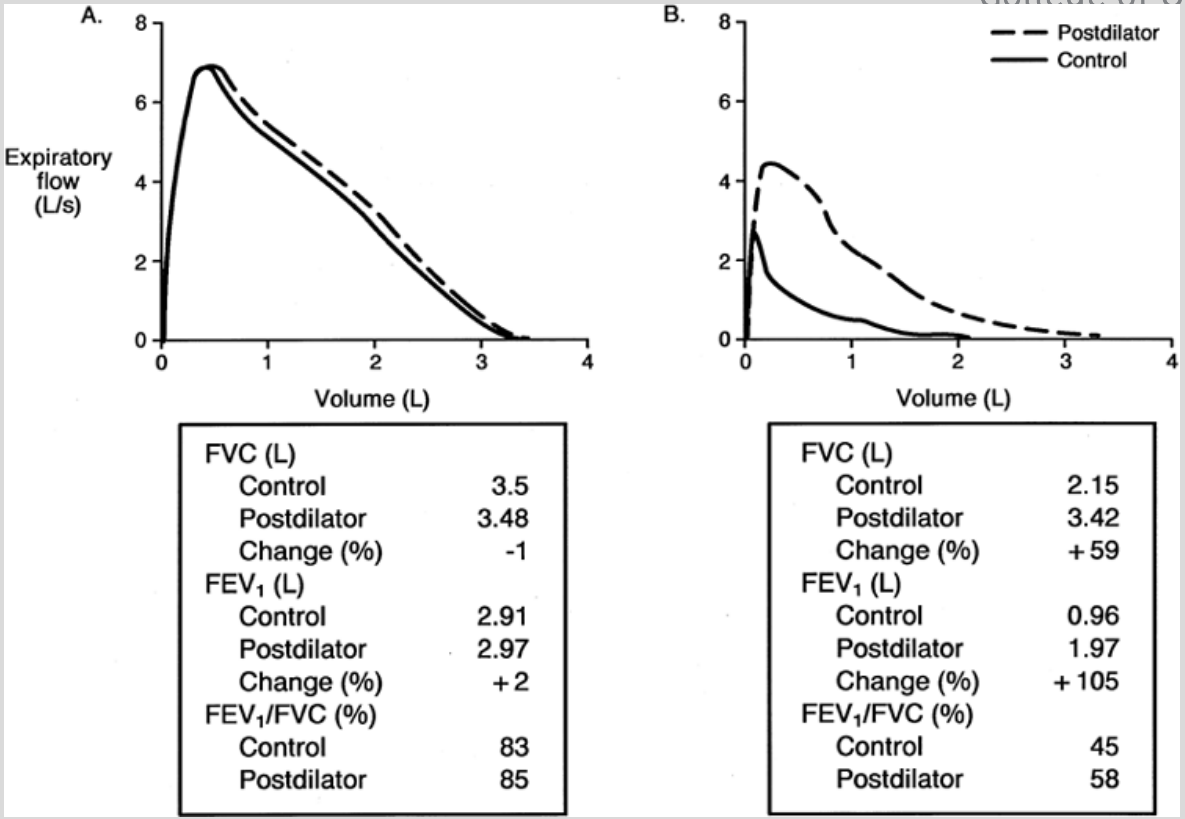


# Bronchodilator Response

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- Degree to which $FEV_1$ improves with inhaled bronchodilator
- Documents *reversible* airflow obstruction
- Significant response if:
  - $FEV_1$ increases by 12% and >200ml
- Request if obstructive pattern on spirometry

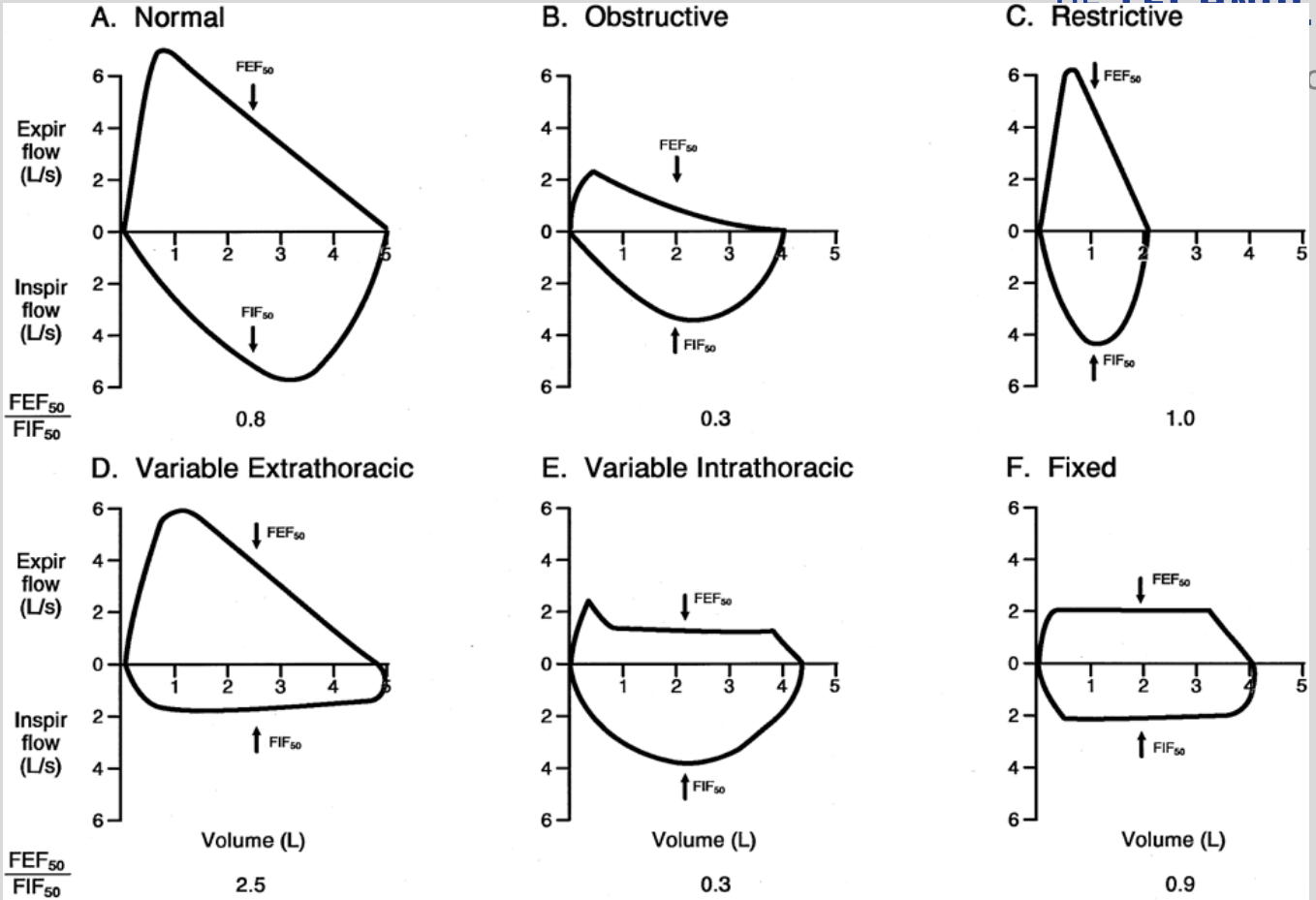
# Differing Responses to Bronchodilator




| | |
|---|---|
| FVC (L) | |
| Control | 3.5 |
| Postdilator | 3.48 |
| Change (%) | -1 |
| $FEV_1$ (L) | |
| Control | 2.91 |
| Postdilator | 2.97 |
| Change (%) | +2 |
| $FEV_1/FVC$ (%) | |
| Control | 83 |
| Postdilator | 85 |

| | |
|---|---|
| FVC (L) | |
| Control | 2.15 |
| Postdilator | 3.42 |
| Change (%) | +59 |
| $FEV_1$ (L) | |
| Control | 0.96 |
| Postdilator | 1.97 |
| Change (%) | +105 |
| $FEV_1/FVC$ (%) | |
| Control | 45 |
| Postdilator | 58 |

# The Major Patterns



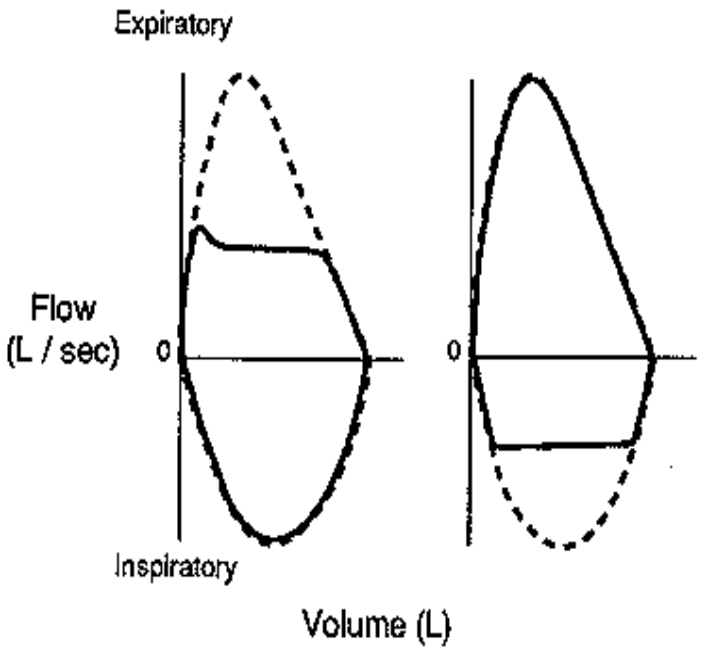
# Upper Airway Obstruction

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

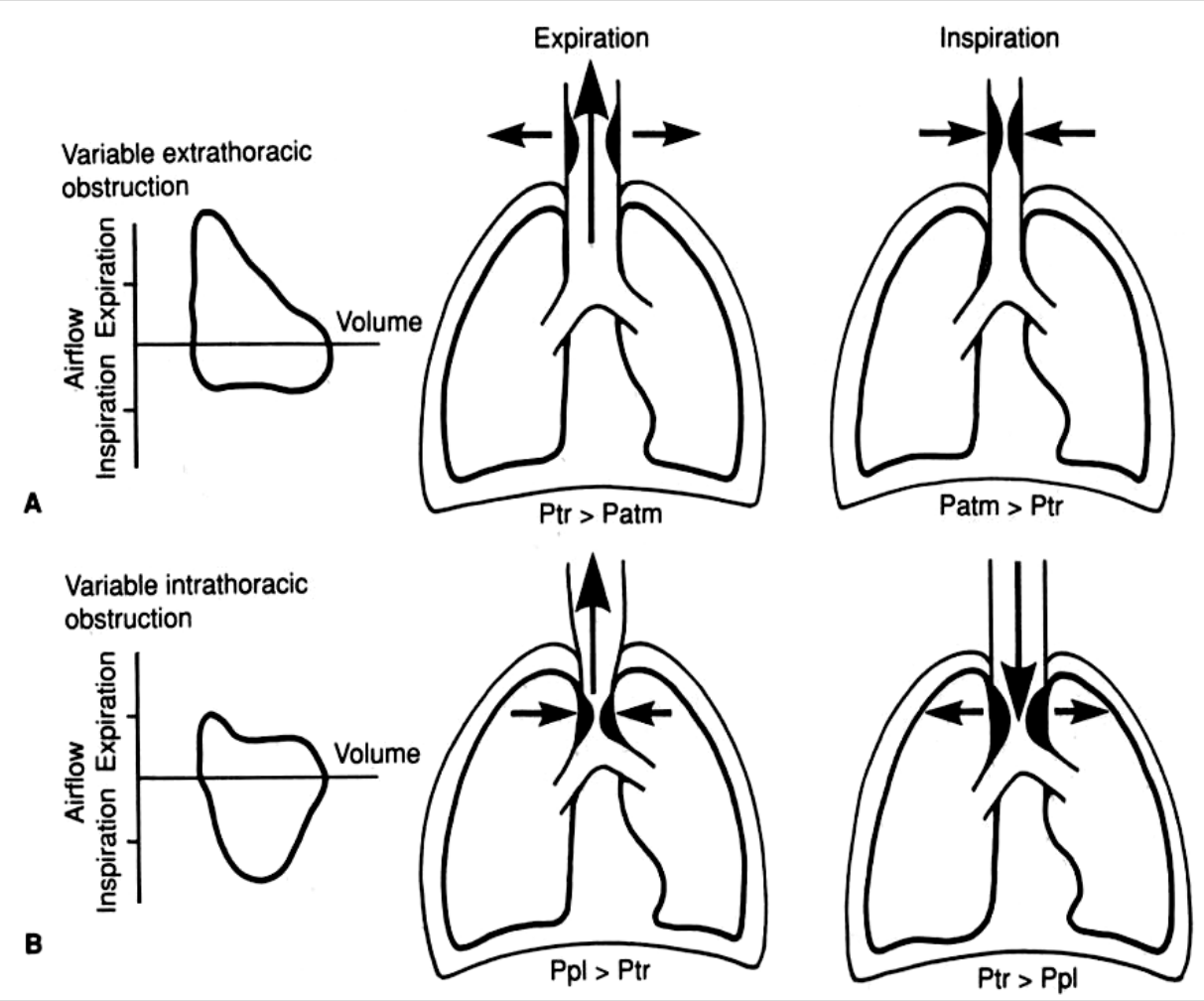
- Variable intrathoracic obstruction
- Variable extrathoracic obstruction
- Fixed obstruction

# Large Airway Obstruction

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine




- Characterized by a truncated inspiratory or expiratory loop


Expiration
Inspiration
Variable extrathoracic obstruction
Airflow
Inspiration Expiration
Volume
A
Ptr > Patm
Patm > Ptr
Variable intrathoracic obstruction
Airflow
Inspiration Expiration
Volume
B
Ppl > Ptr
Ptr > Ppl

# Flow Volume loops in upper airway obstruction: **Variable *Extra*- thoracic**

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

- Edema of epiglottis
- Vocal cord paralysis
- Vocal cord adhesions
- Foreign body

# Flow Volume loops in upper airway obstruction: **Variable *Intra*-Thoracic**

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

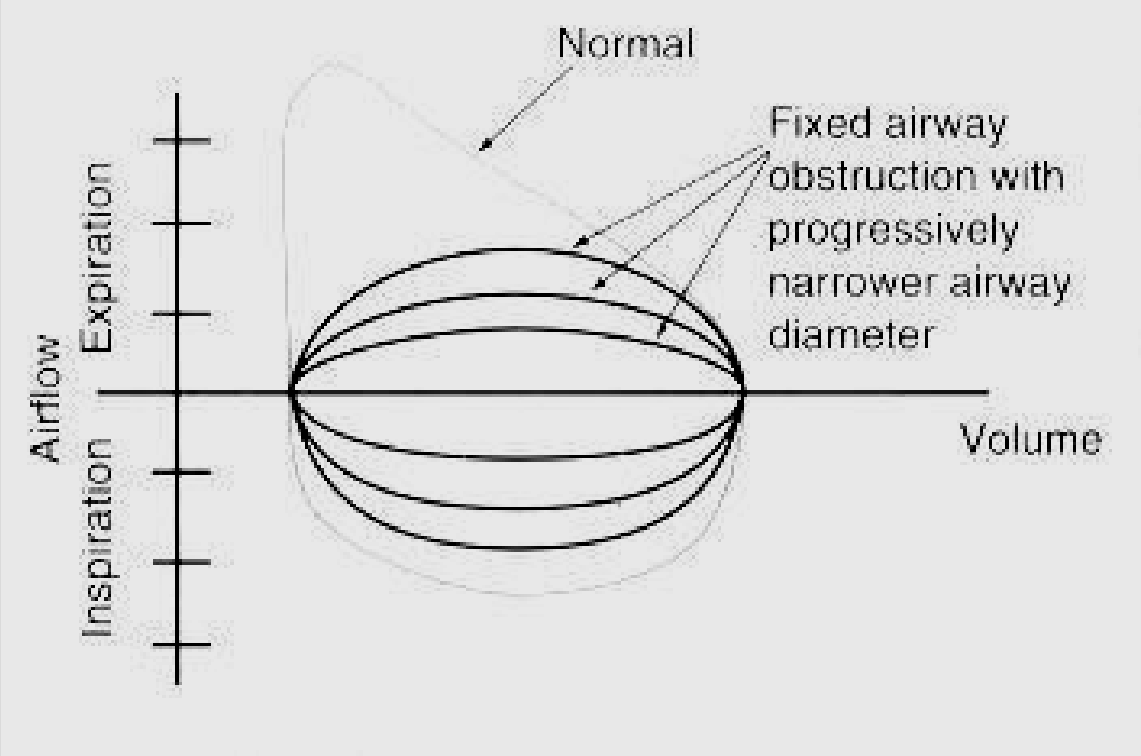
- Polypoid tumor
- Tracheomalacia
- Polychondritis

# Flow Volume loops in upper airway obstruction: ***Fixed***

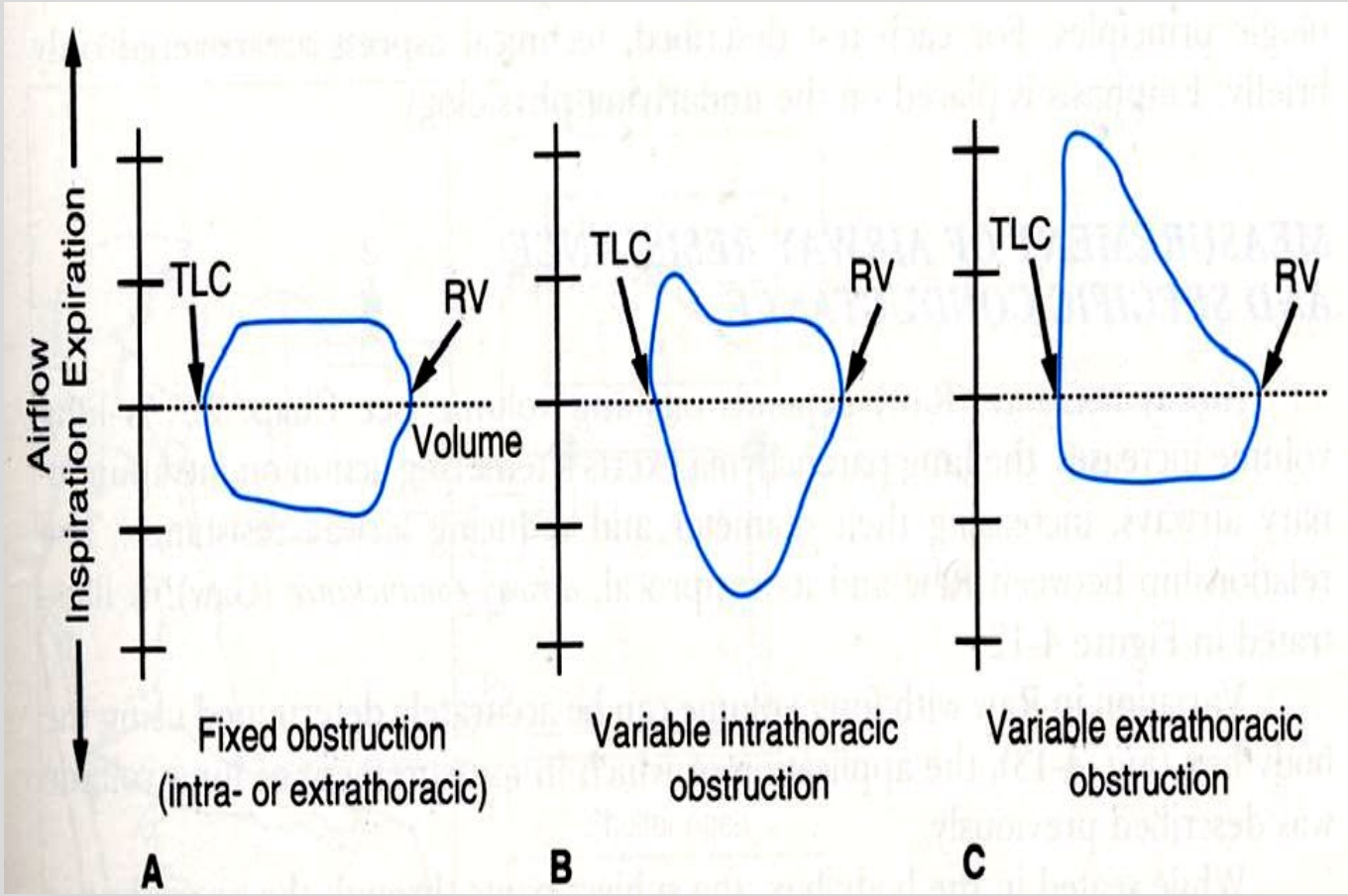
NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

- Tracheal stenosis
- Tracheal goiter
- Hypertrophied tonsils
- Large foreign body


Normal
Fixed airway obstruction with progressively narrower airway diameter
Airflow
Expiration
Inspiration
Volume

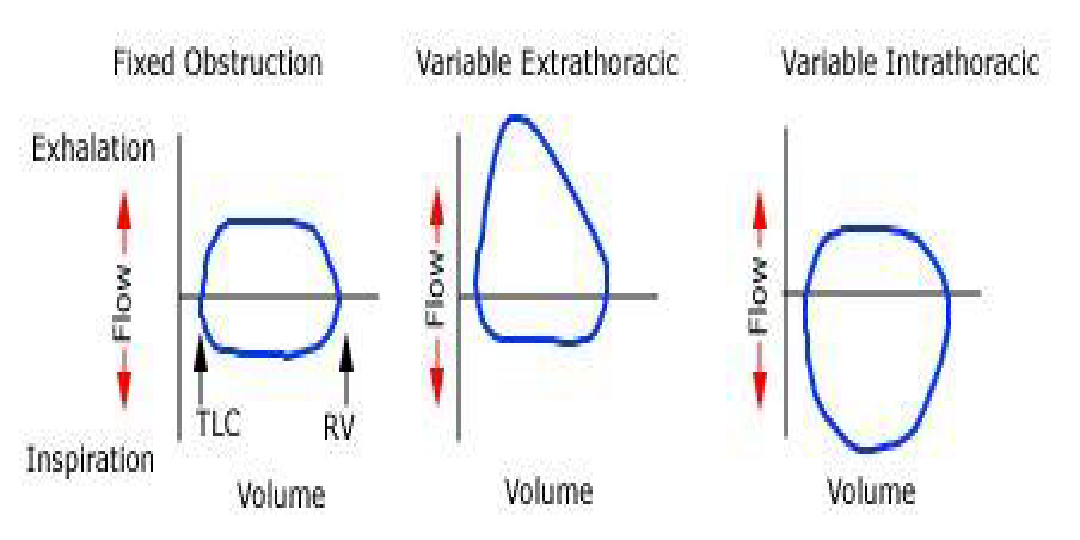
pathic


Airflow
Inspiration Expiration
TLC
RV
Volume
Fixed obstruction
(intra- or extrathoracic)
A
TLC
RV
Variable intrathoracic
obstruction
B
TLC
RV
Variable extrathoracic
obstruction
C

# Upper Airway Obstruction

NEW YORK INSTITUTE OF TECHNOLOGY

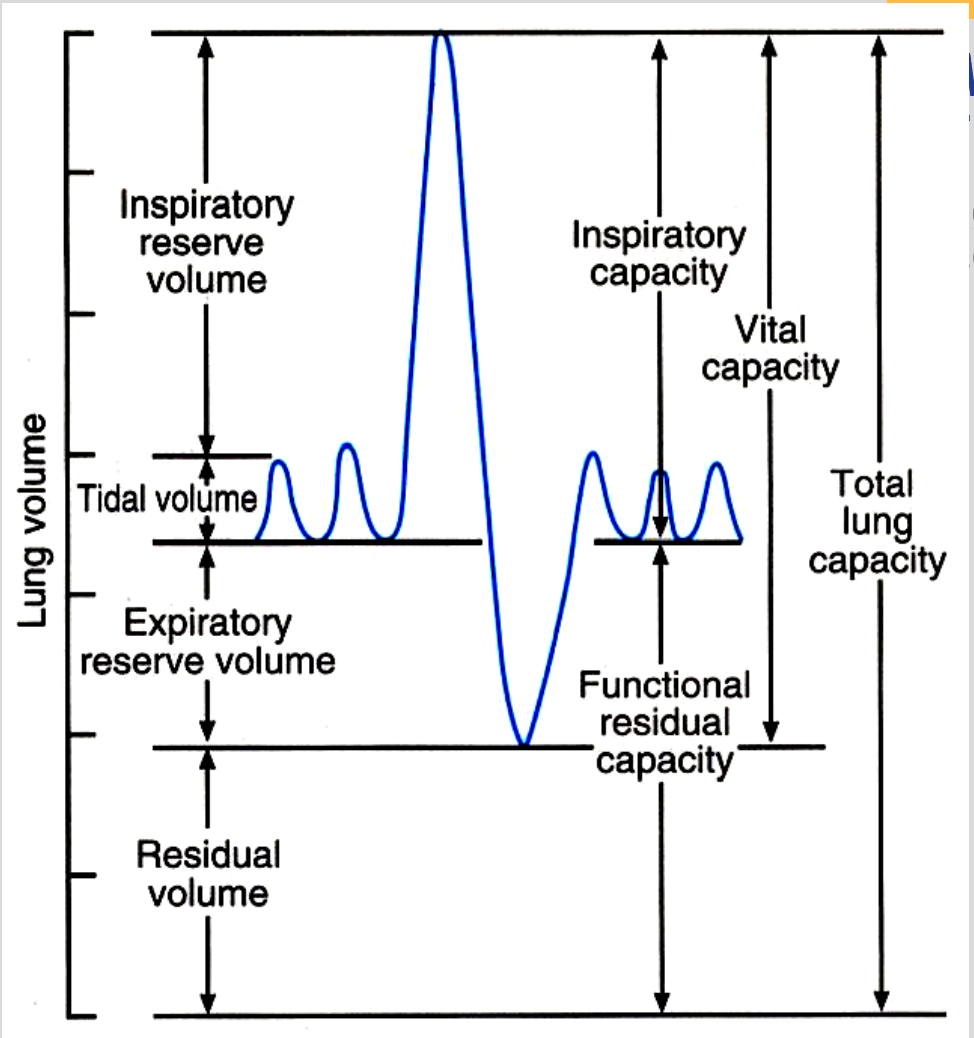
College of Osteopathic Medicine



NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

# LUNG VOLUMES


Lung volume
Inspiratory reserve volume
Tidal volume
Expiratory reserve volume
Residual volume
Inspiratory capacity
Vital capacity
Total lung capacity
Functional residual capacity

# Lung Volumes

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- Measurement:
  - helium dilution
  - nitrogen washout
  - body plethysmography
- Indications:
  - Diagnose restrictive component
  - Differentiate types of obstructive lung dz

# Body Plethysmograph

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic






# Lung Volumes

- IRV= Inspiratory Reserve volume
- $V_T$= Tidal Volume
- TLC= Total Lung Capacity
- RV= Residual Volume
- ERV= Expiratory Reserve Volume
- IC = Inspiratory Capacity

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

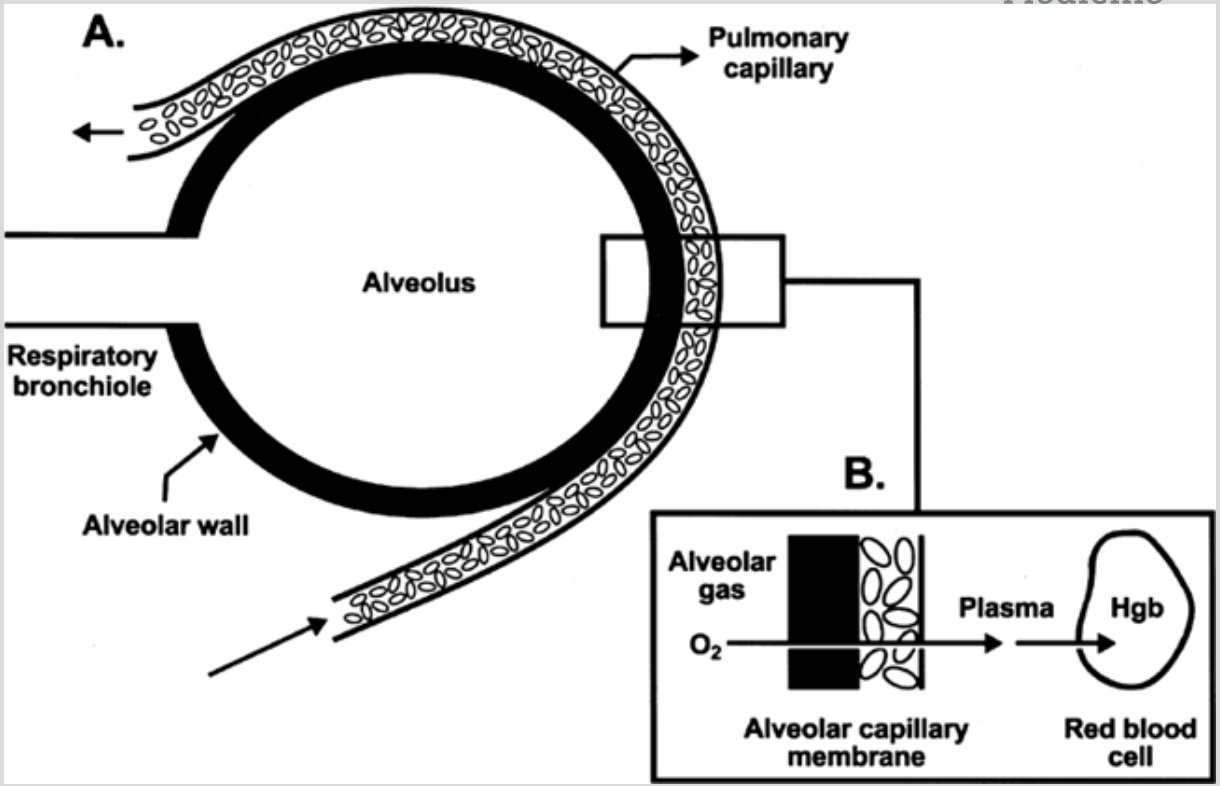
# Lung Volumes – Patterns

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- Obstructive
  - **TLC > 120% predicted**
  - RV > 120% predicted
- Restrictive
  - **TLC < 80% predicted**
  - RV < 80% predicted

# Diffusing Capacity

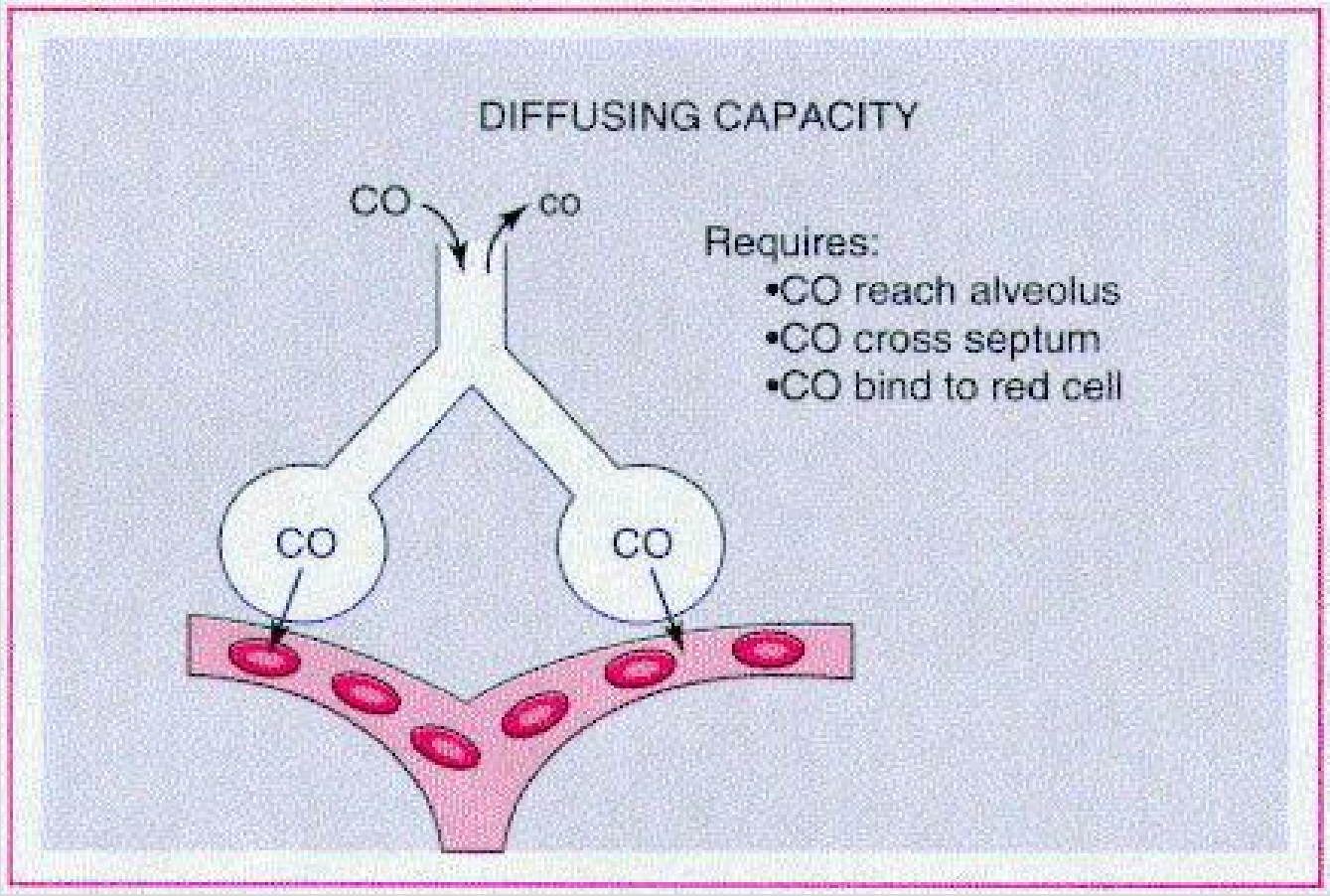
College of Osteopathic Medicine

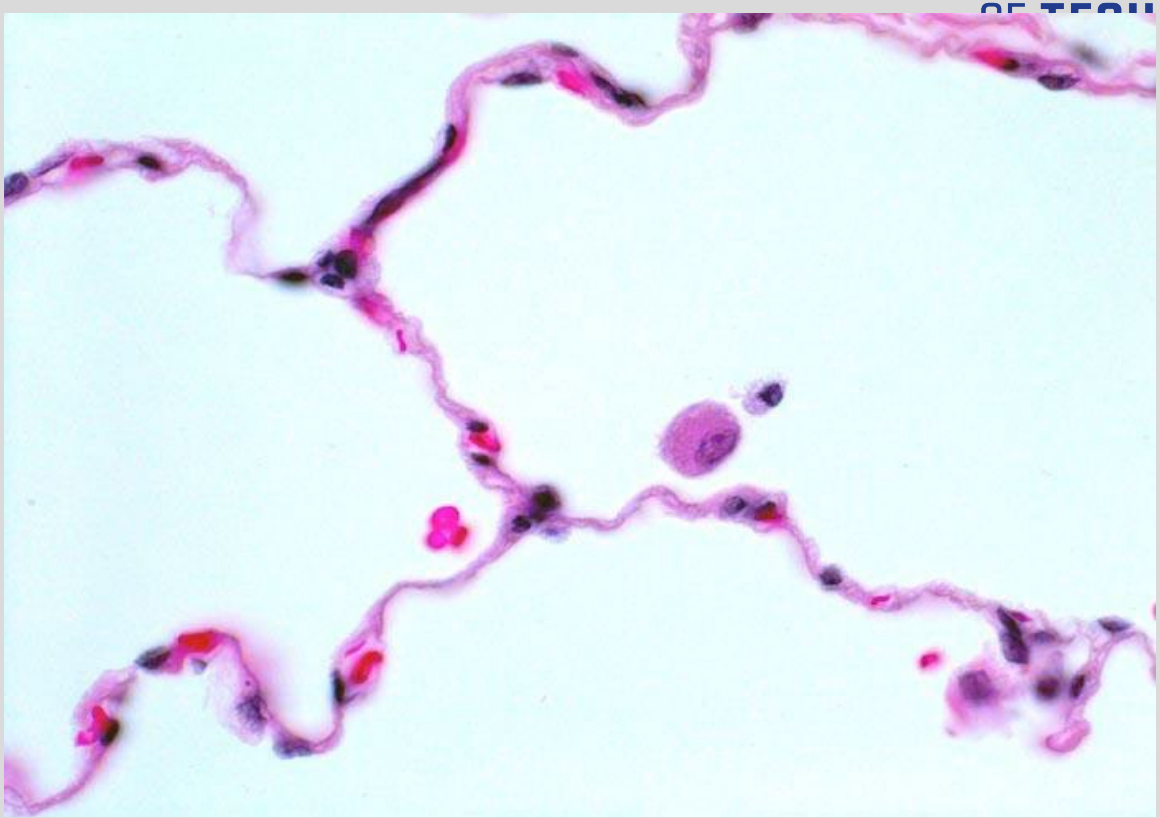


# Diffusing Capacity Determined by:

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- Surface area of alveoli
- Thickness of alveolar membrane
- Volume of blood in capillary bed
- Reaction rate of the test gas with Hgb
  - We use **DLCO** – carbon monoxide


DIFFUSING CAPACITY
CO
co
Requires:
•CO reach alveolus
•CO cross septum
•CO bind to red cell
CO
CO

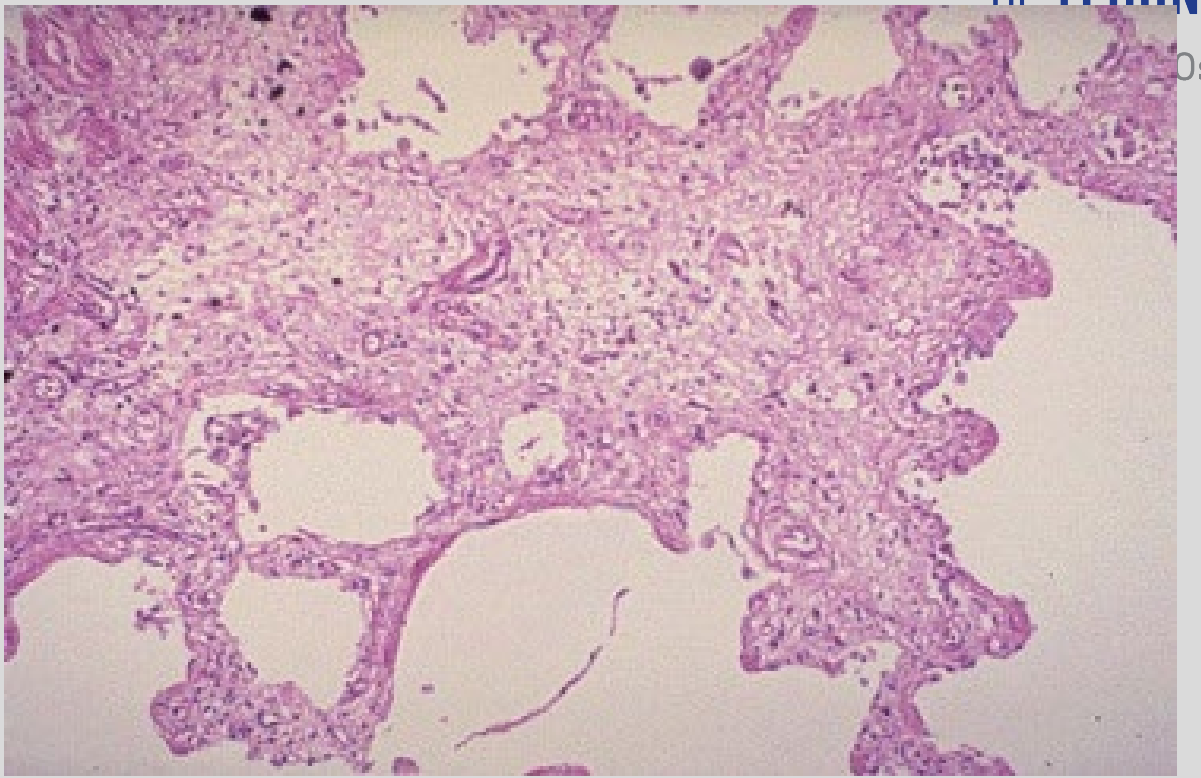


# Diffusing Capacity

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic

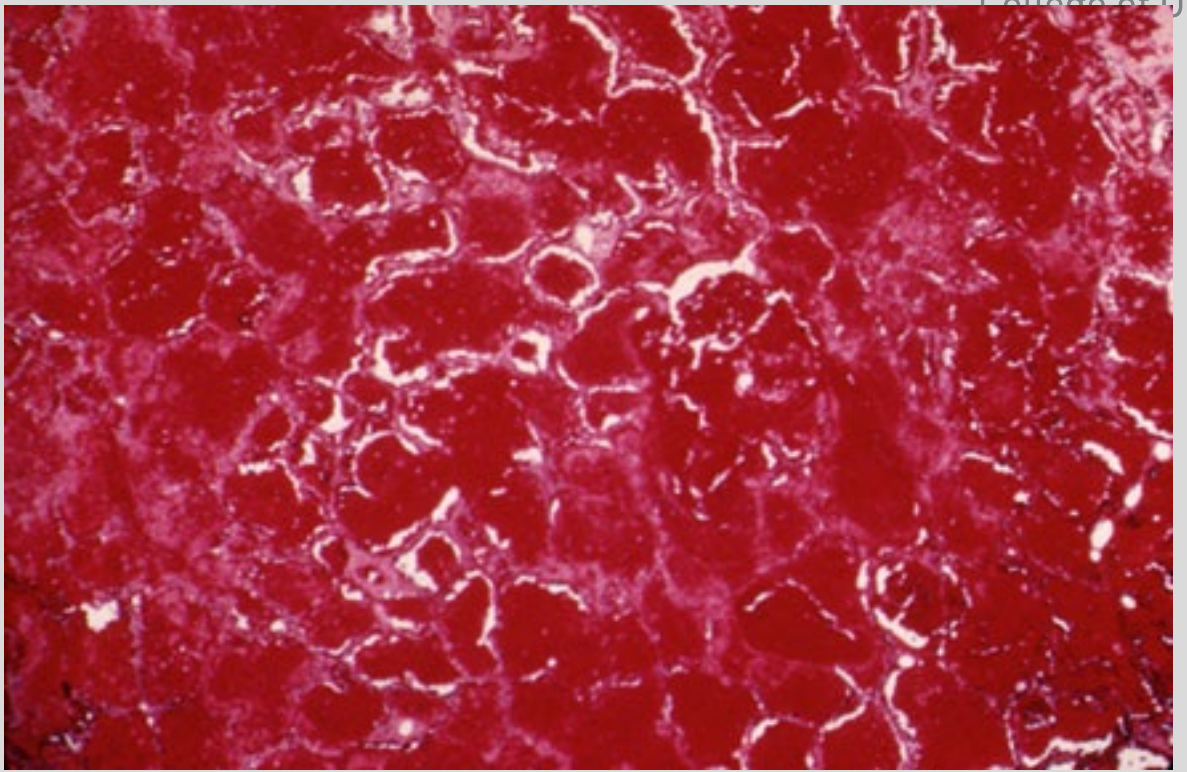
- **Decreased DLCO**
  (<80% predicted)
  - Obstructive lung disease
    - Emphysema
  - Parenchymal disease
    - Interstitial (fibrosis)
  - Pulmonary vascular disease (pulm HTN)
  - Anemia

- **Increased DLCO**
  (>120-140% predicted)
  - Asthma (or normal)
  - Pulmonary hemorrhage
  - Polycythemia
  - Left to right shunt



NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic



# DLCO — Indications

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- Differentiate asthma from COPD
- Evaluation and severity of restrictive lung disease
- Early stages of pulmonary hypertension

# PFT Patterns

INSTITUTE
LOGY
College of Osteopathic

- **Emphysema**
  - $FEV_1/FVC$ <70%
  - "Scooped" FV curve
  - TLC *increased*
  - Increased compliance
  - DLCO *decreased*
- **Chronic Bronchitis**
  - $FEV_1/FVC$ <70%
  - "Scooped" FV curve
  - TLC *normal*
  - Normal compliance
  - DLCO usually *normal*

# Bronchoprovocation

W YORK INSTITUTE
TECHNOLOGY
ege of Osteopathic
Medicine

- Useful for diagnosis of asthma in the setting of *normal* pulmonary function tests
- Common agents:
  - Methacholine, Histamine, Mannitol
- Diagnostic if: ≥20% decrease in $FEV_1$

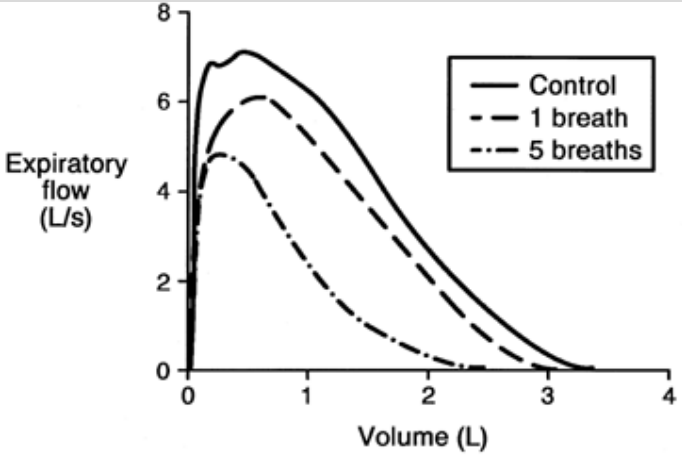
# Methacholine Challenge

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

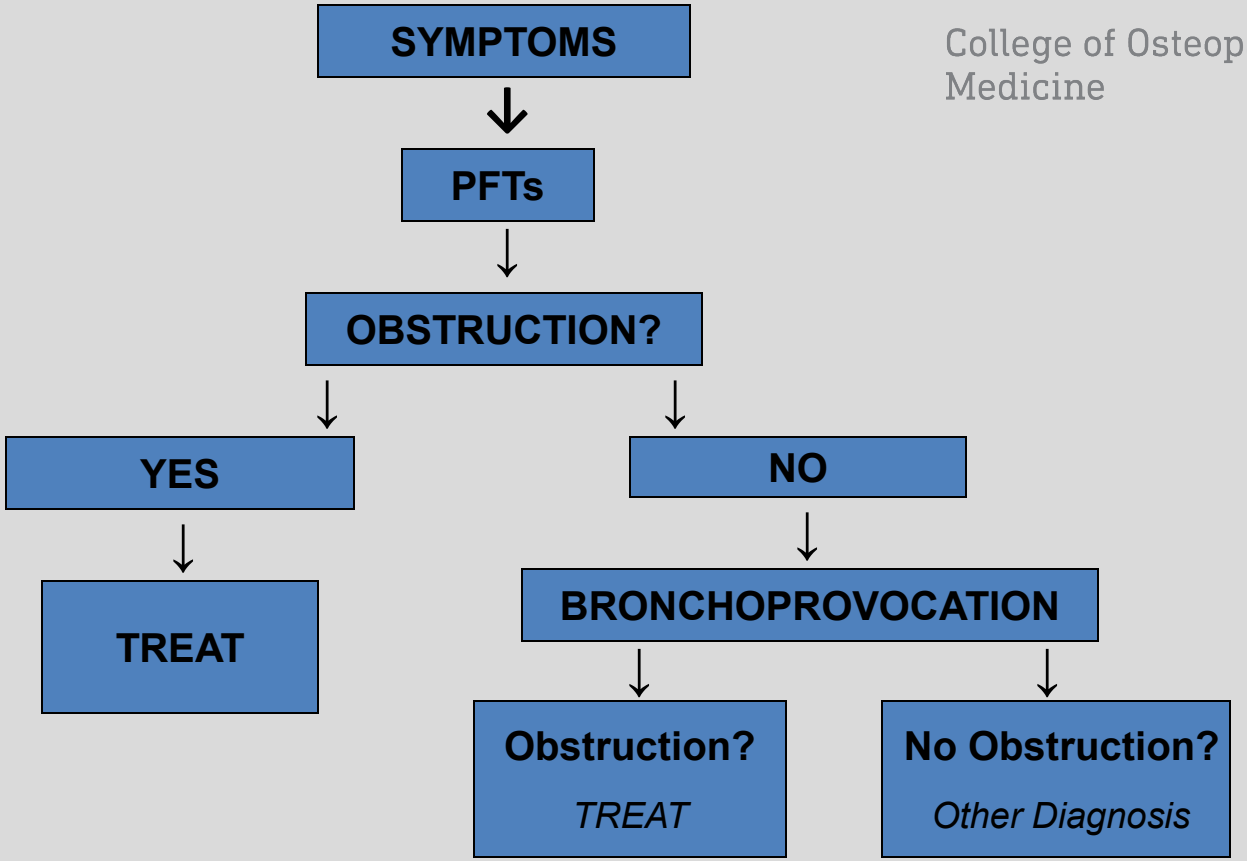
- Inhaled methacholine
  - Cholinergic drug that stimulates muscarinic receptors
  - Causes contraction of smooth muscle in airway
  - Degree to which muscle reacts depends on its reactivity
- Predictive of asthma attacks by cold air, exercise, pollens, infection, etc.
- Given up to 5 doses

# Positive Methacholine

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic




| | Control | 1 breath | 5 breaths |
|---|---|---|---|
| FVC (L) | 3.49 | 3.1 | 2.47 |
| $FEV_1$ (L) | 2.86 | 2.54 | 1.82 |
| $FEV_1$ (% decrease) | | 11 | 36 |
| $FEV_1$/FVC (%) | 82 | 82 | 74 |


SYMPTOMS
PFTs
OBSTRUCTION?
YES
NO
TREAT
BRONCHOPROVOCATION
Obstruction?
TREAT
No Obstruction?
Other Diagnosis

opathic

**TABLE 4-1.** *COMMON PATTERNS OF PULMONARY FUNCTION ABNORMALTIES*

| MEASUREMENT | RESTRICTIVE DISORDERS | OBSTRUCTIVE DISORDERS | |
|---|---|---|---|
| | | Mild | Severe |
| FVC | ↓ ↓ | Normal | ↓ |
| $FEV_1$ | ↓ ↓ | ↓ | ↓ ↓ |
| $FEV_1$/FVC% | Normal | ↓ | ↓ ↓ |
| $FEF_{25\%-75\%}$ | ↓ ↓ | ↓ | ↓ ↓ |
| FRC | ↓ | Normal | ↑ |
| RV | ↓ | Normal | ↑ |
| TLC | ↓ | Normal | ↑ |

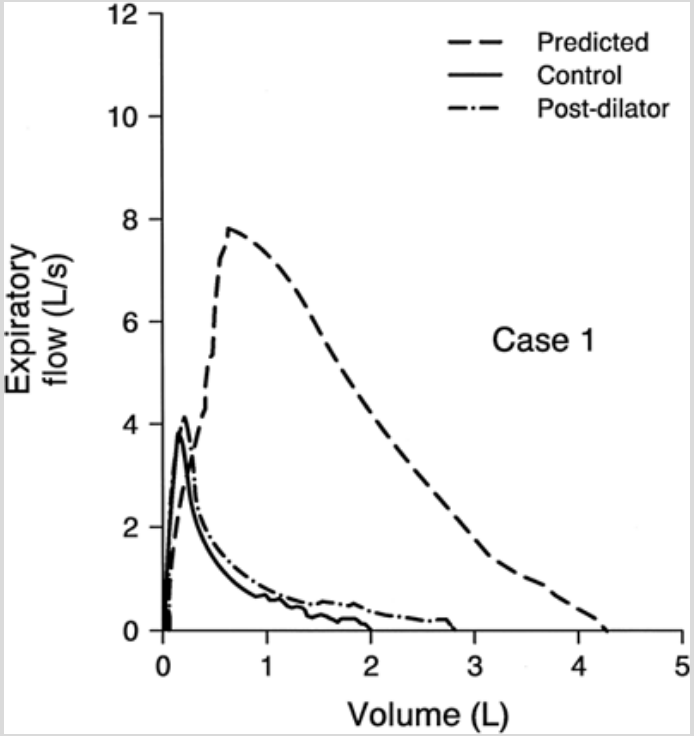
NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

# CASES

# Smoked 74 p-y progressive sob

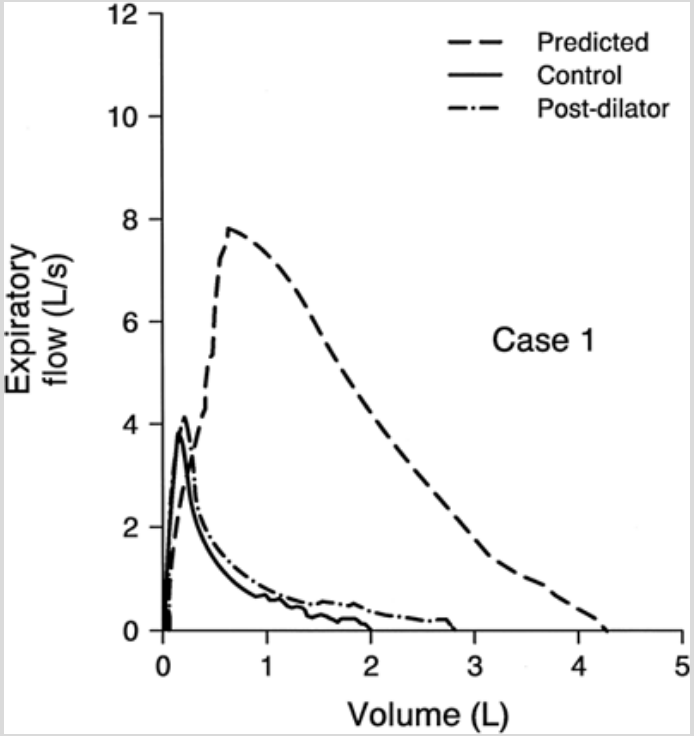
YORK INSTITUTE
CHNOLOGY
College of Osteopathic
Medicine




- ⇩ FEV1 ratio
- FEV1 - 31%

# Smoked 74 p-y progressive sob

College of Osteopathic Medicine

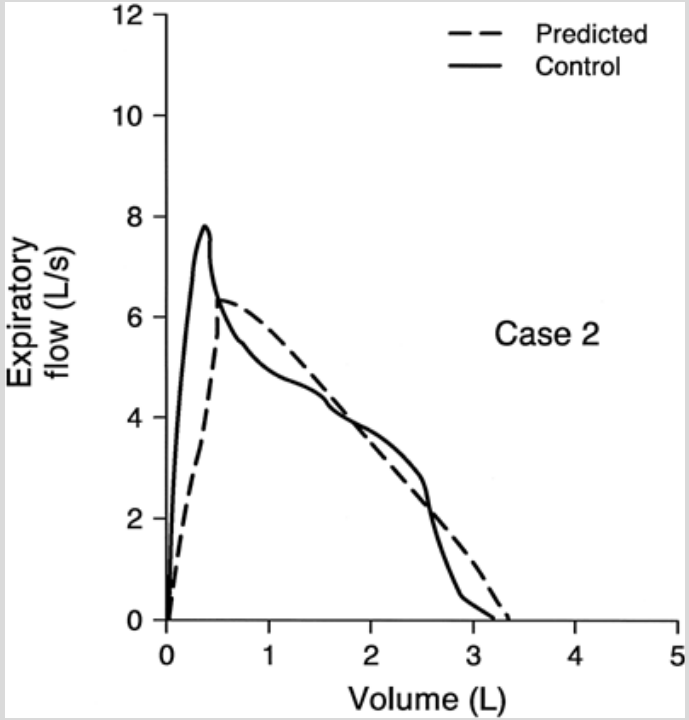



- ↓ FEV1 ratio
- FEV1 - 31%

- **Severe obstruction**

# 36 y o female

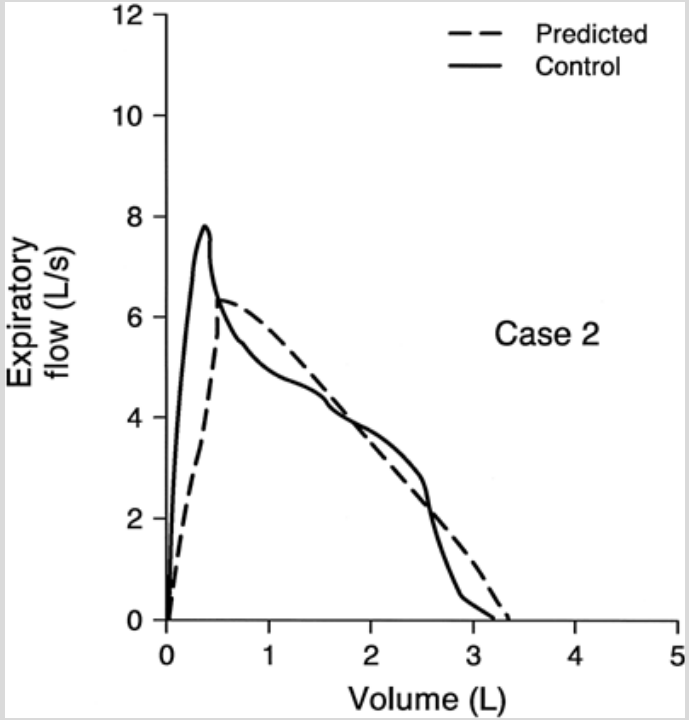
College of Osteopathic Medicine




- Nrml FEV1 ratio
- Nrml FEV1
- Nrml TLC

# 36 y o female

College of Osteopathic Medicine

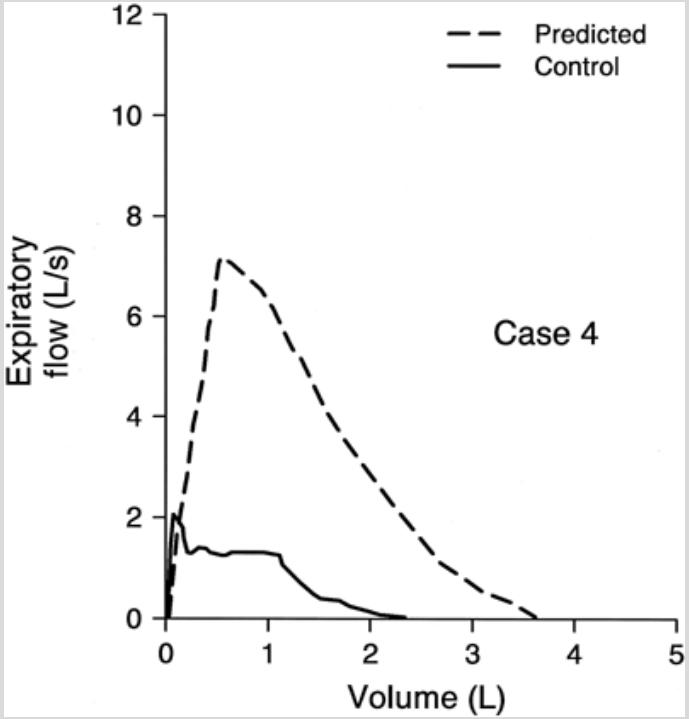



- Nrml FEV1 ratio
- Nrml FEV1
- Nrml TLC

- **NORMAL**
- Slight “knee” pattern
- benign

# 76 y o male - sob

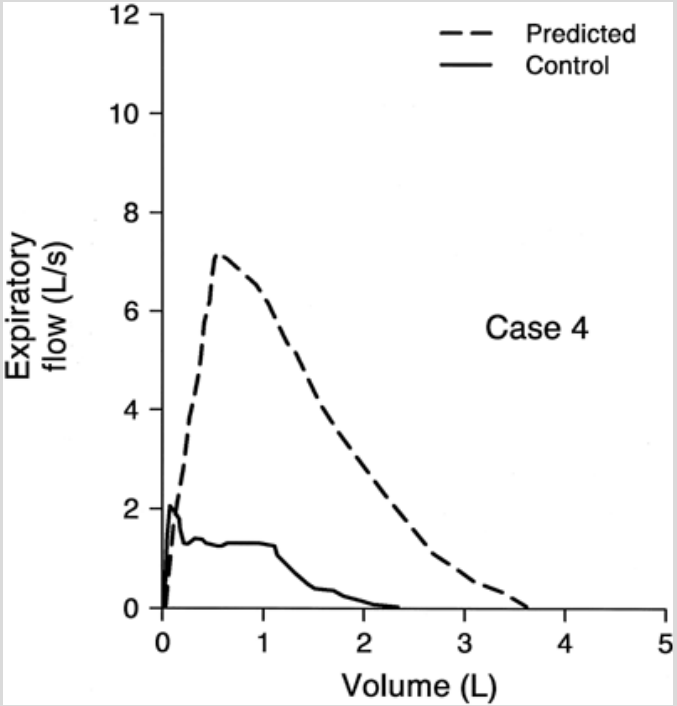
College of Osteopathic Medicine

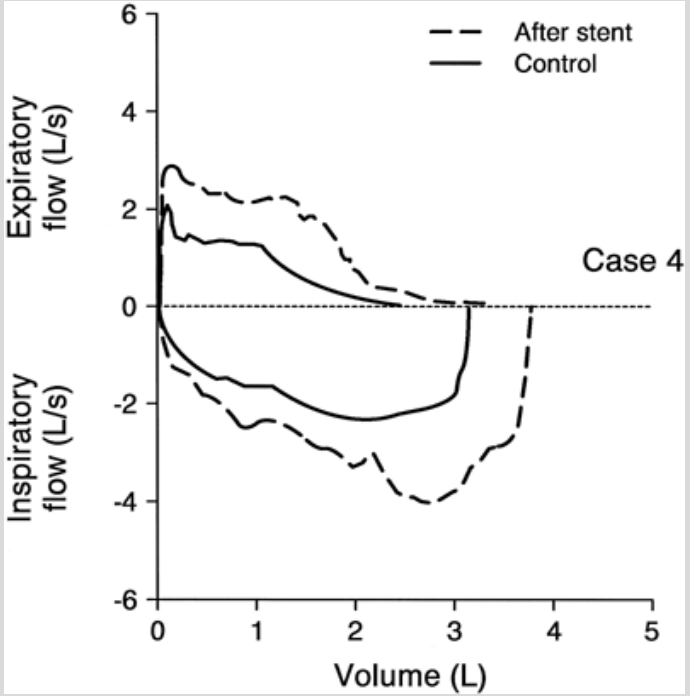



- Decreased FEV1 ratio
- FEV1 --- 47%
- But….Look at that loop….
- Flattened??

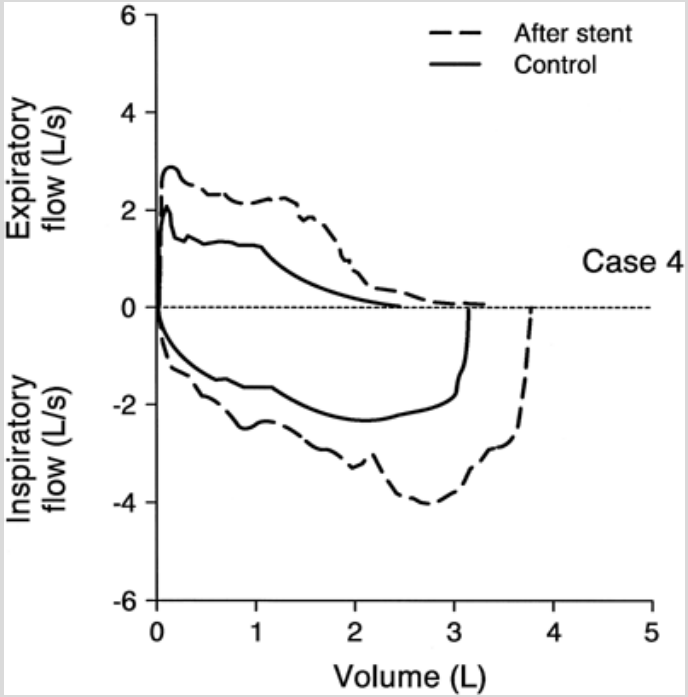
# The rest of the loop!!!

College of Osteopathic Medicine



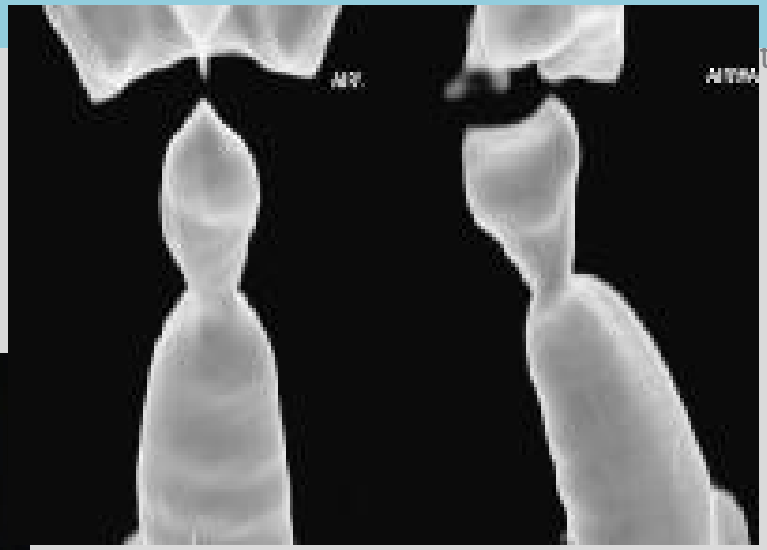
# 76 y o male - sob




- Decreased FEV1 ratio
- FEV1 47%
- But….Look at that loop….
- Flattened??

- **ANCA-associated Vasculitis**
- **Subglottic stenosis**

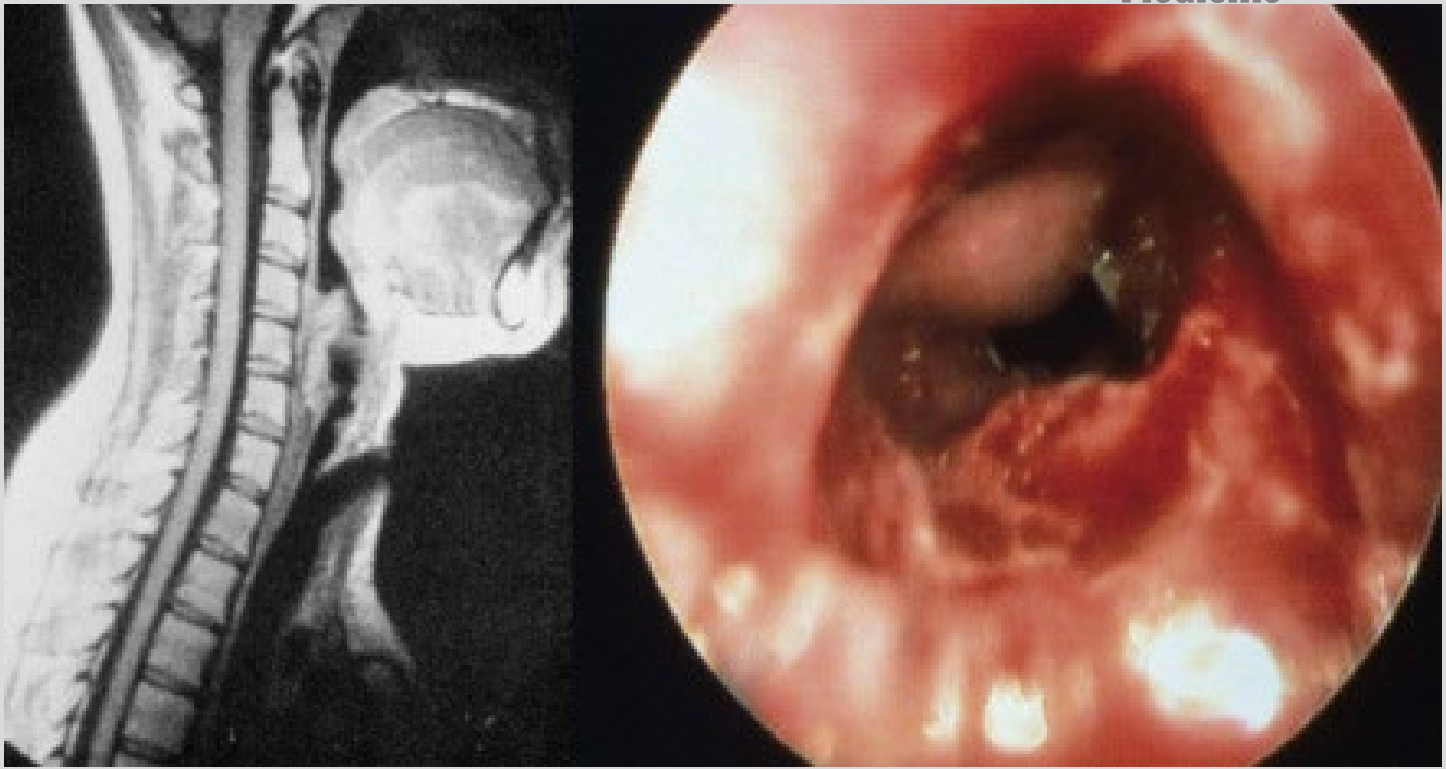
# Subglottic Stenosis



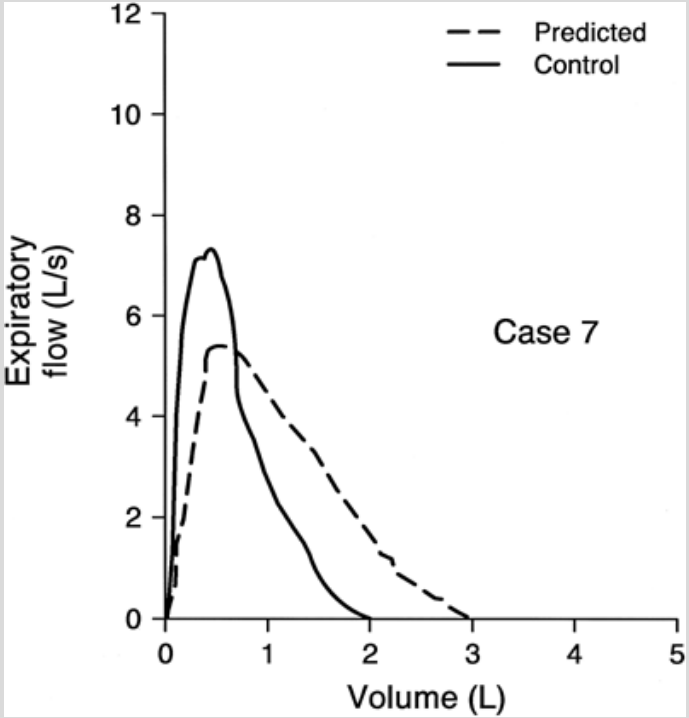


# Subglottic Stenosis

College of Osteopathic Medicine

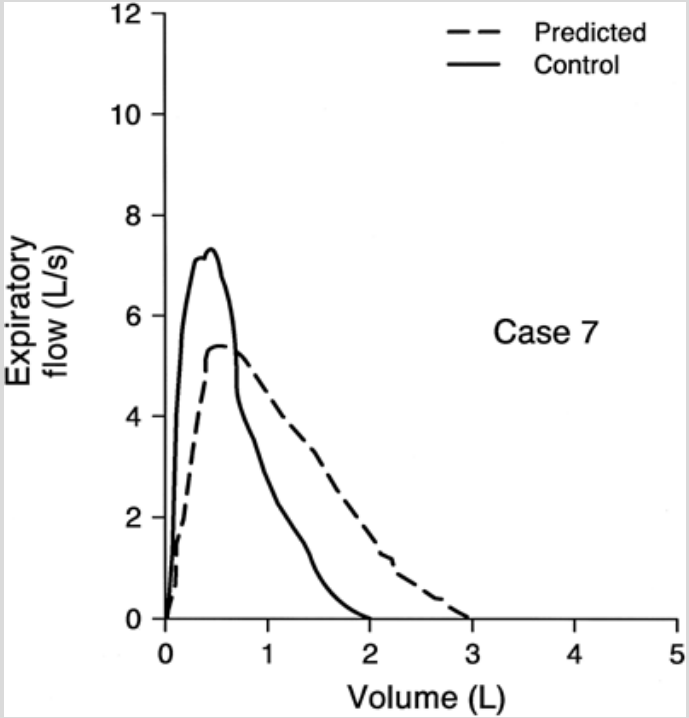


# 67 nonsmoker c occasional sob and wheezing; obese




- Nrml FEV1 ratio
- FEV1 --- 72%
- Nrml TLC

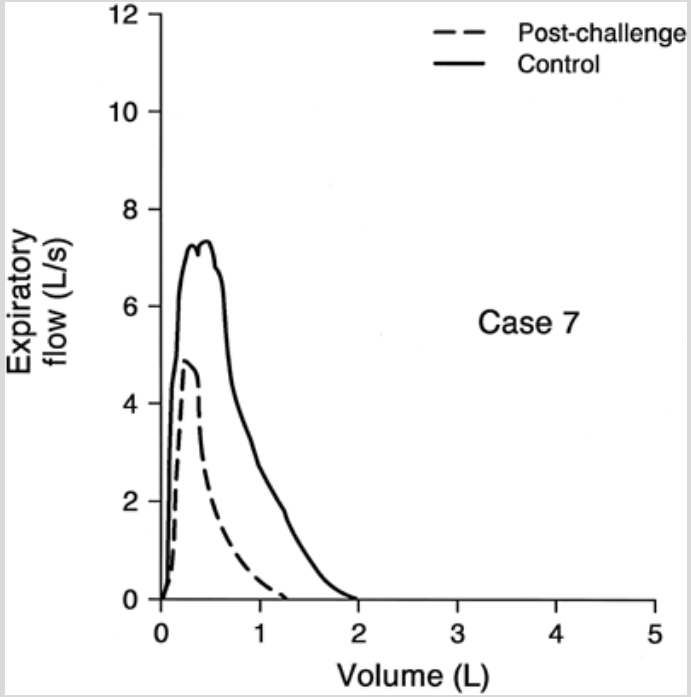
# 67 nonsmoker c occasional sob and wheezing;  obese




- Nrml FEV1 ratio
- FEV1 72%
- Nrml TLC
- Looks restricted…
- Mild scoop..
- But wheezing, sob, so…..

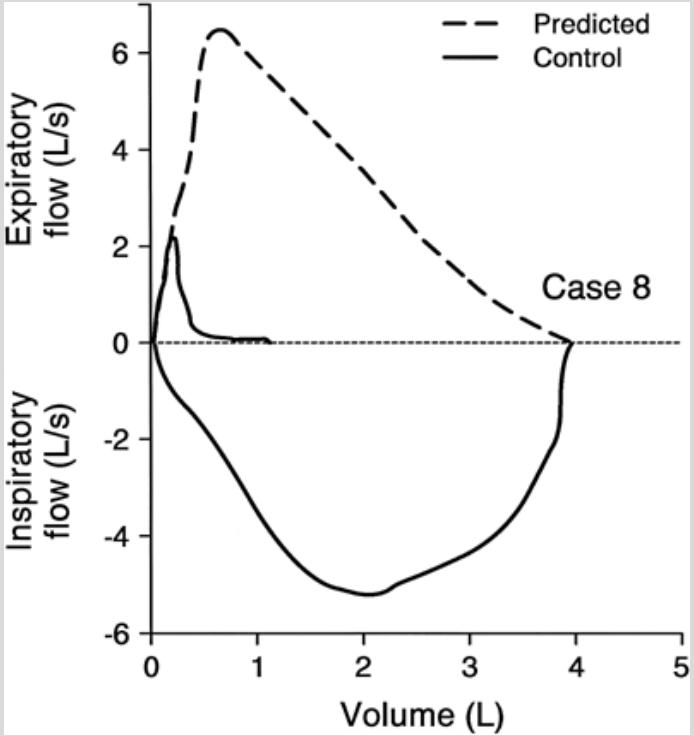
# Methacholine Challenge!!!

College of Osteopathic Medicine




- Post-challenge 41% decrease in FEV1
- Treated c inhaled bronchodilator

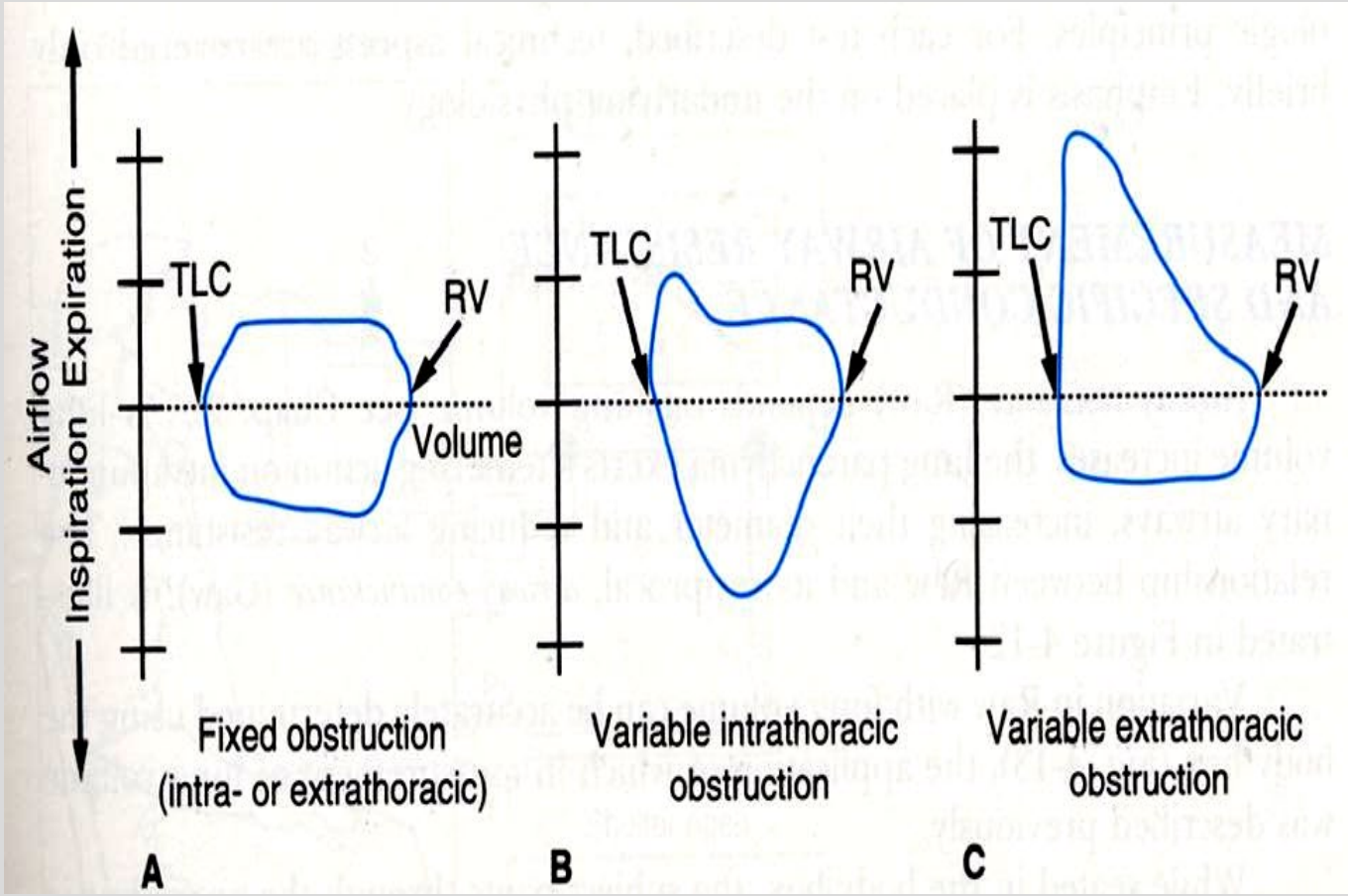
# 43 F 16 p-y smoker recent wheezing, CP, SOB




- Decreased FEV1 ratio
- FEV1 17%

- Look at the loops….

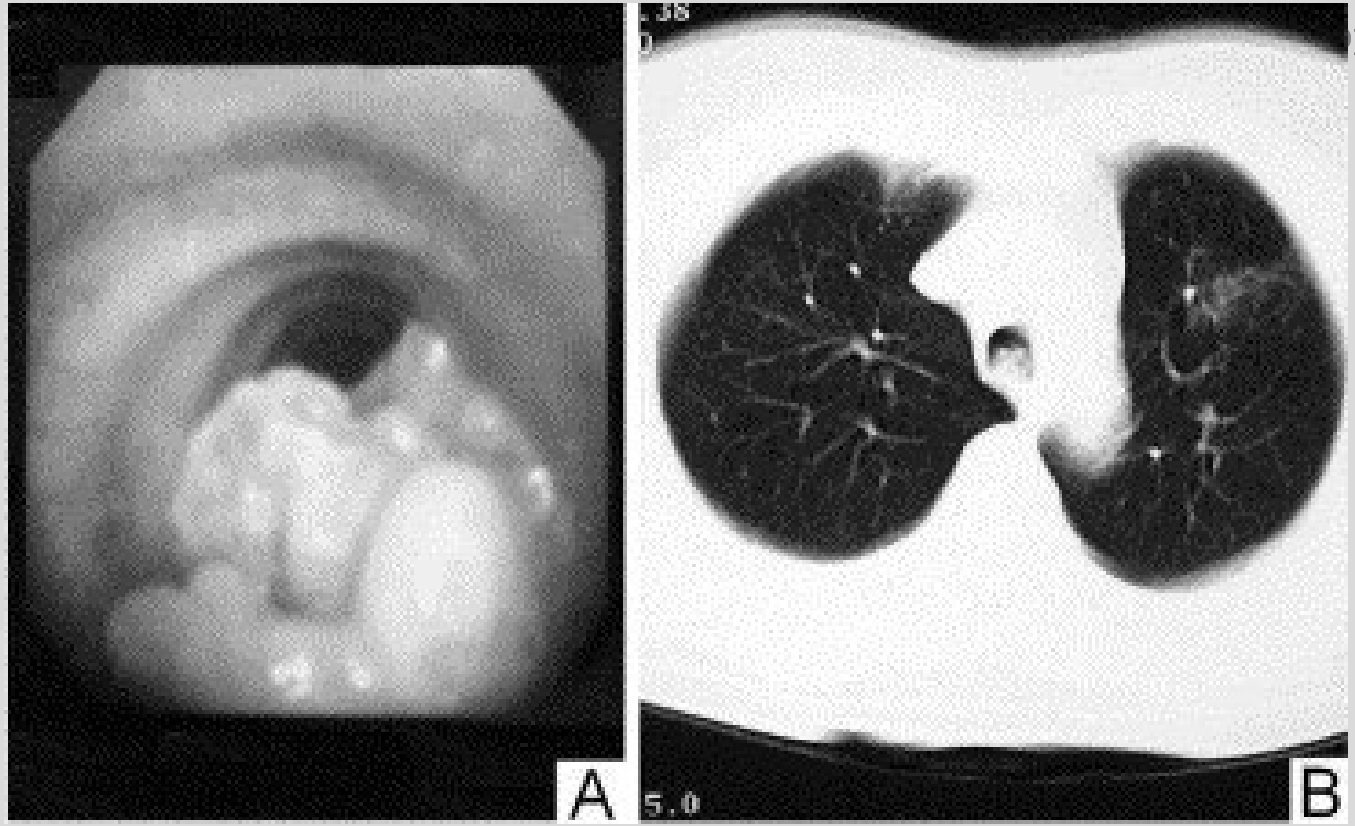
pathic


Airflow
Inspiration Expiration
TLC
RV
Volume
Fixed obstruction
(intra- or extrathoracic)
A
TLC
RV
Variable intrathoracic
obstruction
B
TLC
RV
Variable extrathoracic
obstruction
C

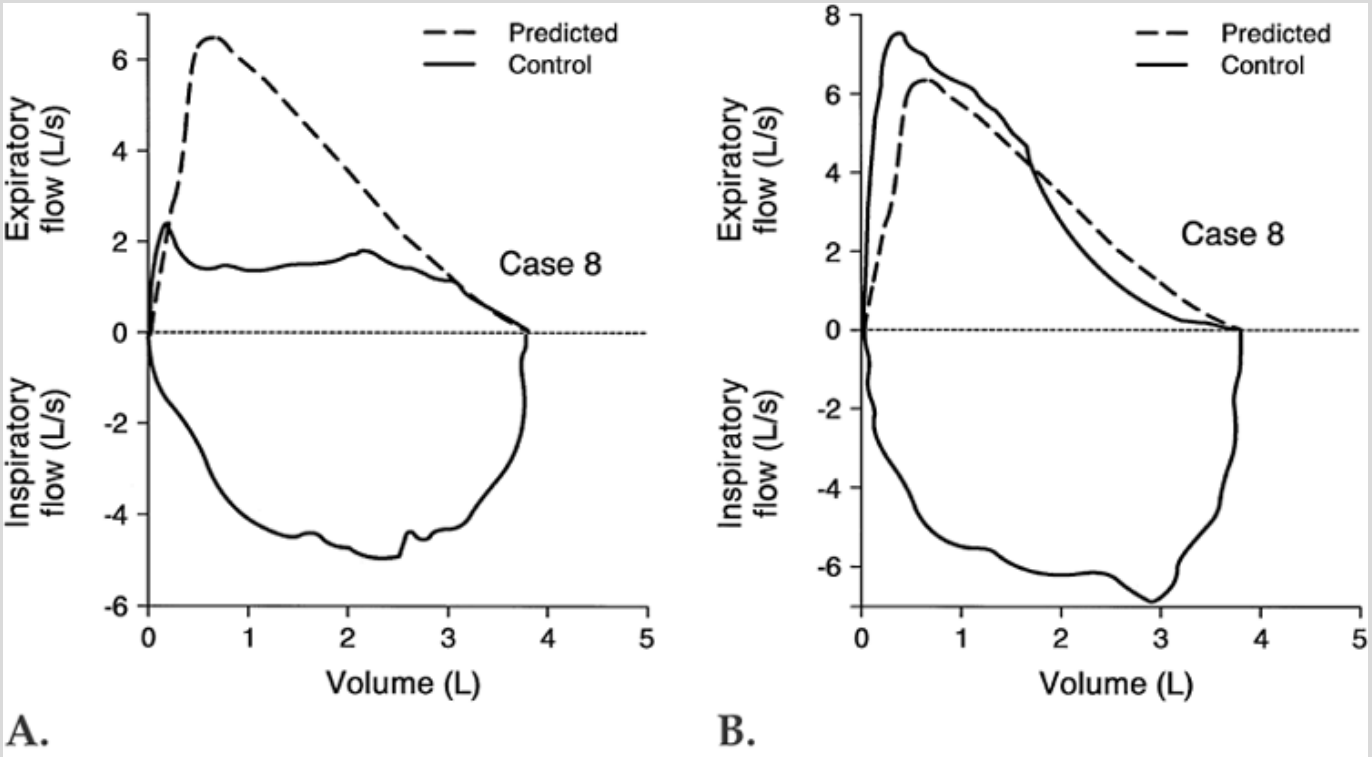
# Squamous Cell Lung Ca

INSTITUTE
LOGY

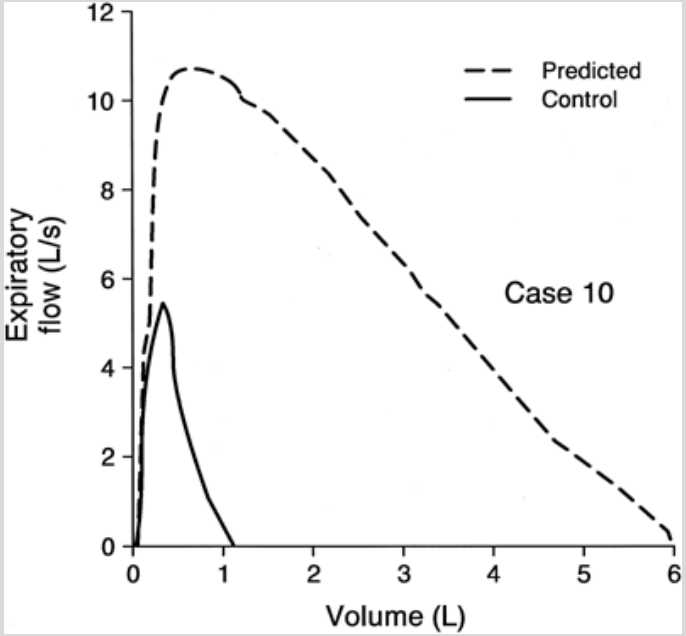
pathic



# post Laser tx (A), then post chemo/rdx (B)

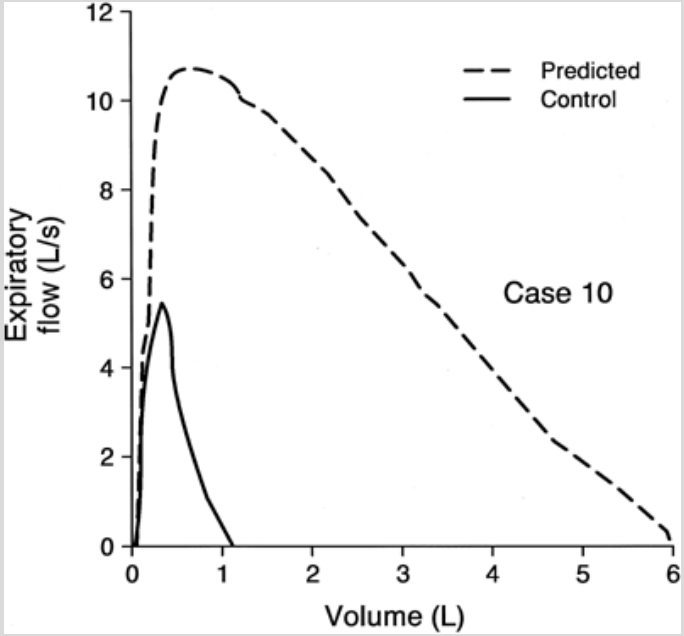


# 60 M SOB c exertion non-smoker

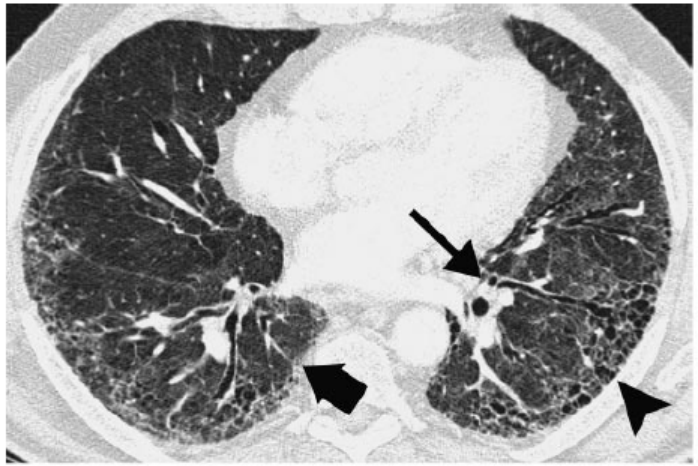



- FEV1 ratio Nrml
- TLC -- 28%
- FEV1 -- 21%
- FVC – 19%

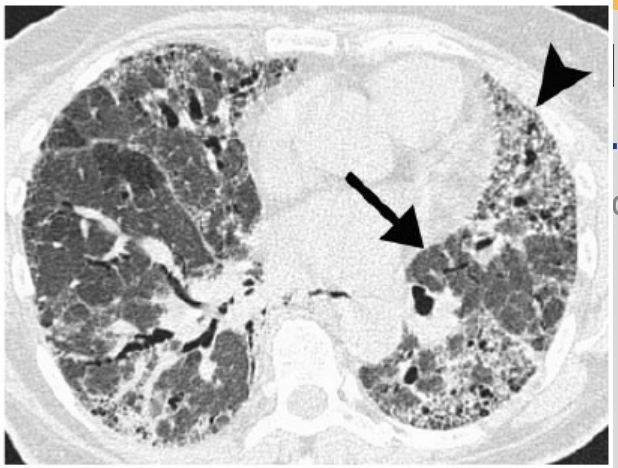
# 60 M SOB c exertion non-smoker



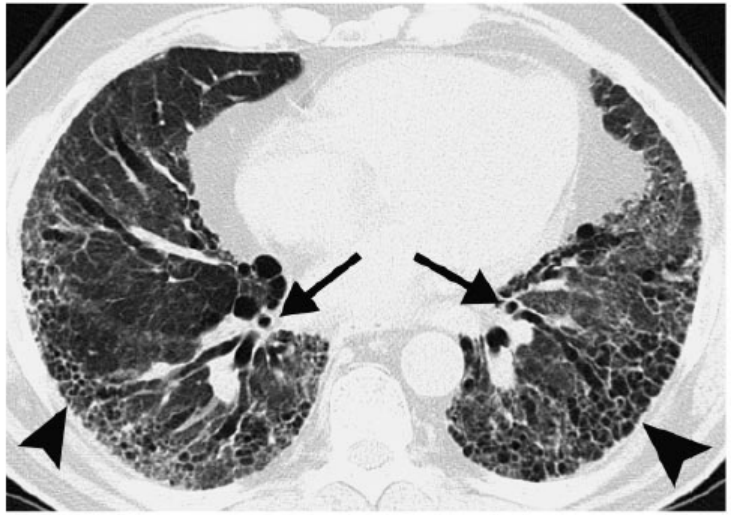

- FEV1 ratio Nrml
- TLC -- 28%
- FEV1 -- 21%
- FVC – 19%
- **Severe restrictive lung disease**

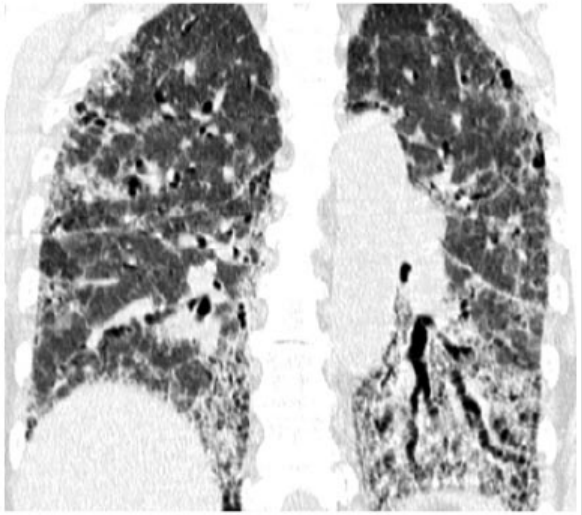


4a.



5a.

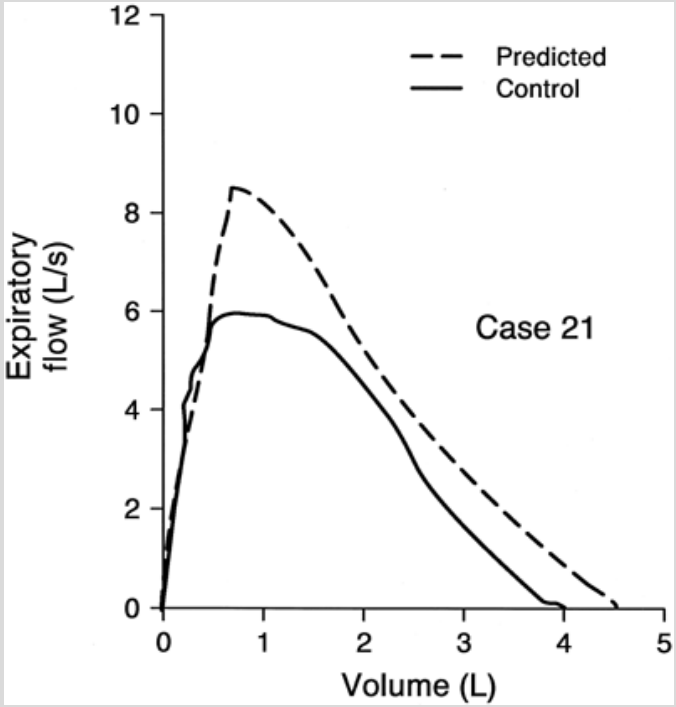




INSTITUTE
OGY

opathic

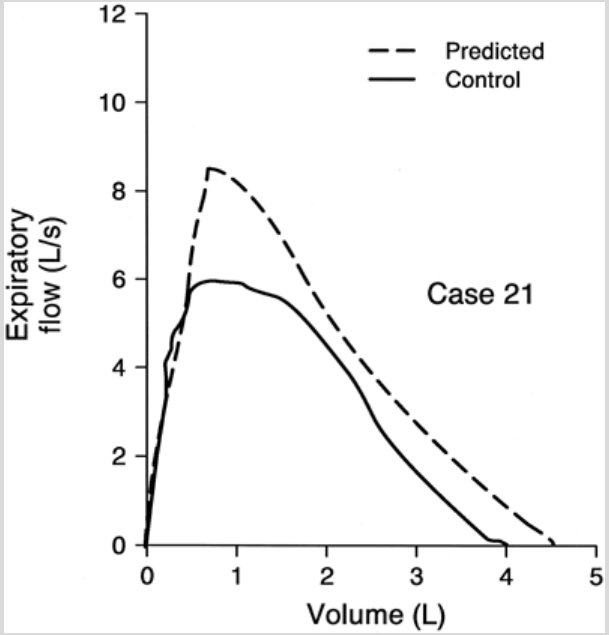
# 50 M – DOE nonsmoker
# h/o bulbar poliomyelitis

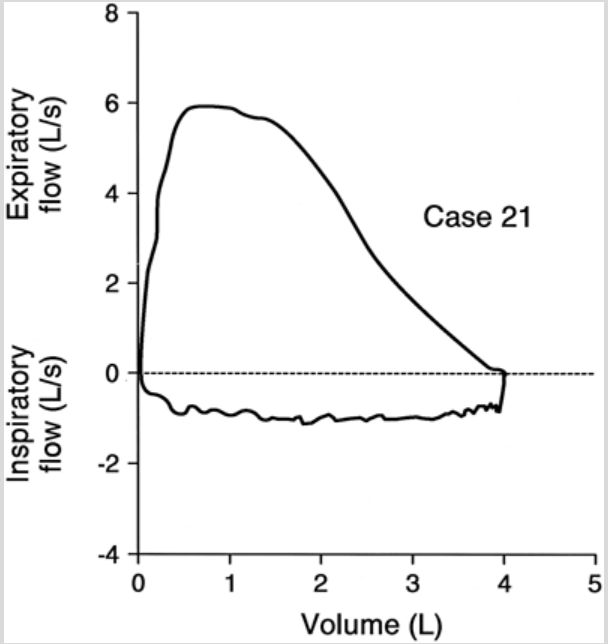



- Nrml FEV1 ratio
- Nrml FEV1

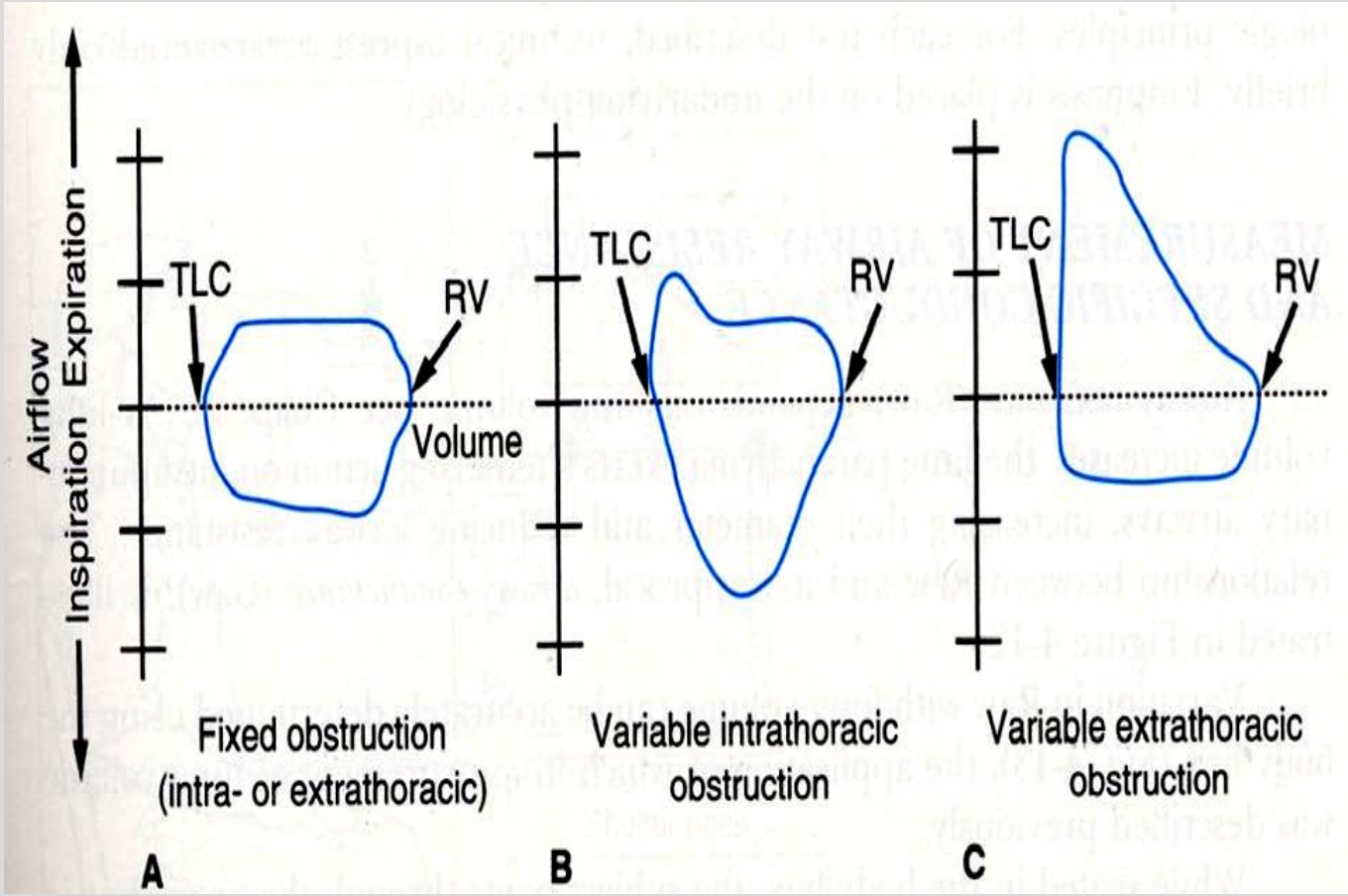
- Where is rest of curve???

# 50 M – DOE nonsmoker
# h/o bulbar poliomyelitis

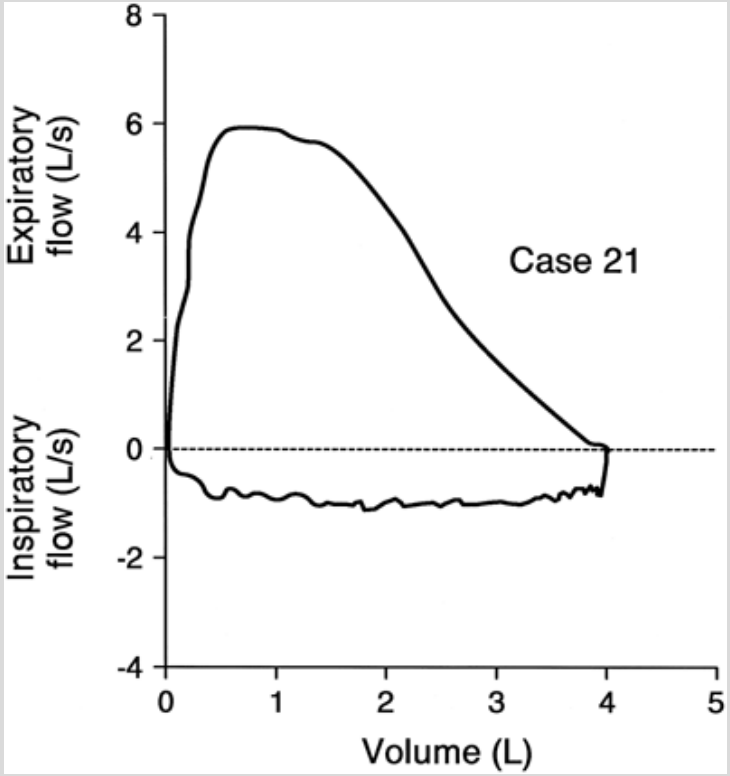


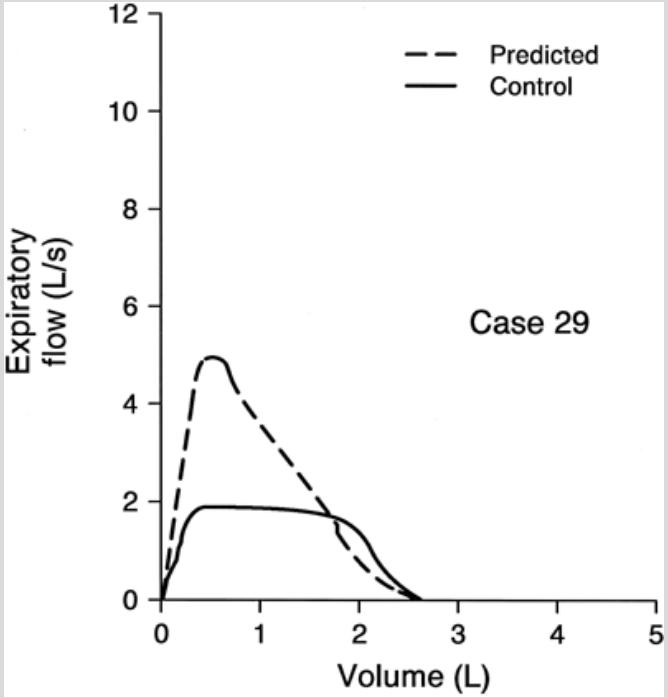
pathic


Airflow
Inspiration Expiration
TLC
RV
Volume
Fixed obstruction
(intra- or extrathoracic)
A
TLC
RV
Variable intrathoracic
obstruction
B
TLC
RV
Variable extrathoracic
obstruction
C

# R Vocal cord paralysis
# L partial vocal cord paralysis



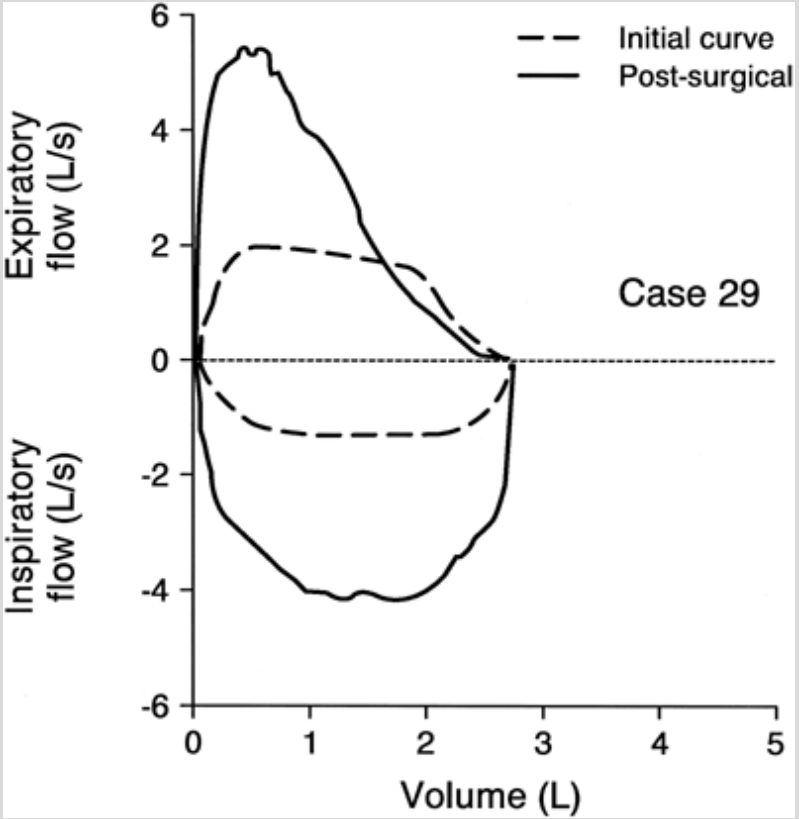
# 64 F nonsmoker gradual onset DOE over 5 years, noisy breathing



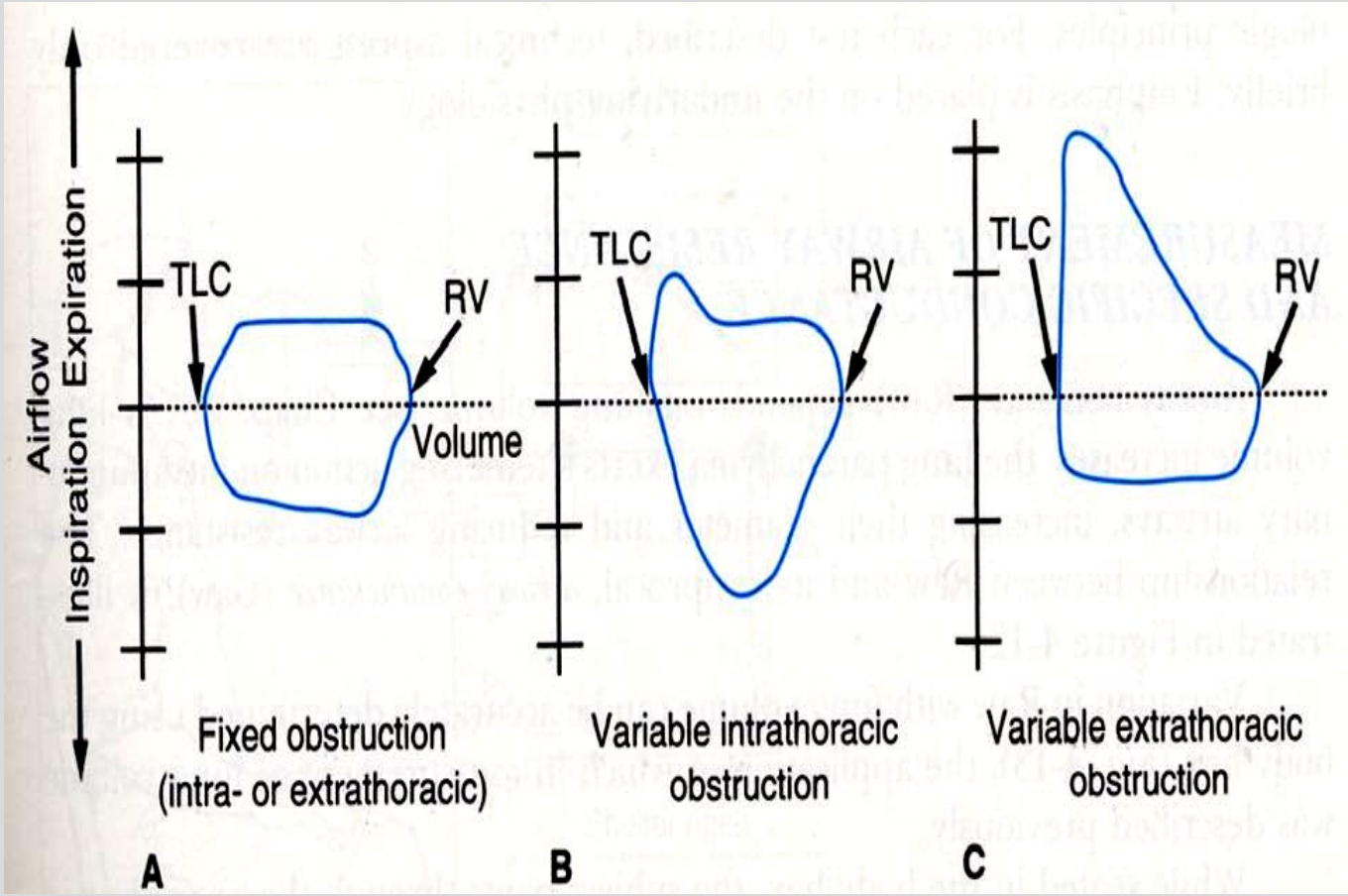

- Where is the inspiratory loop???

# Subglottic stenosis (and post-repair)

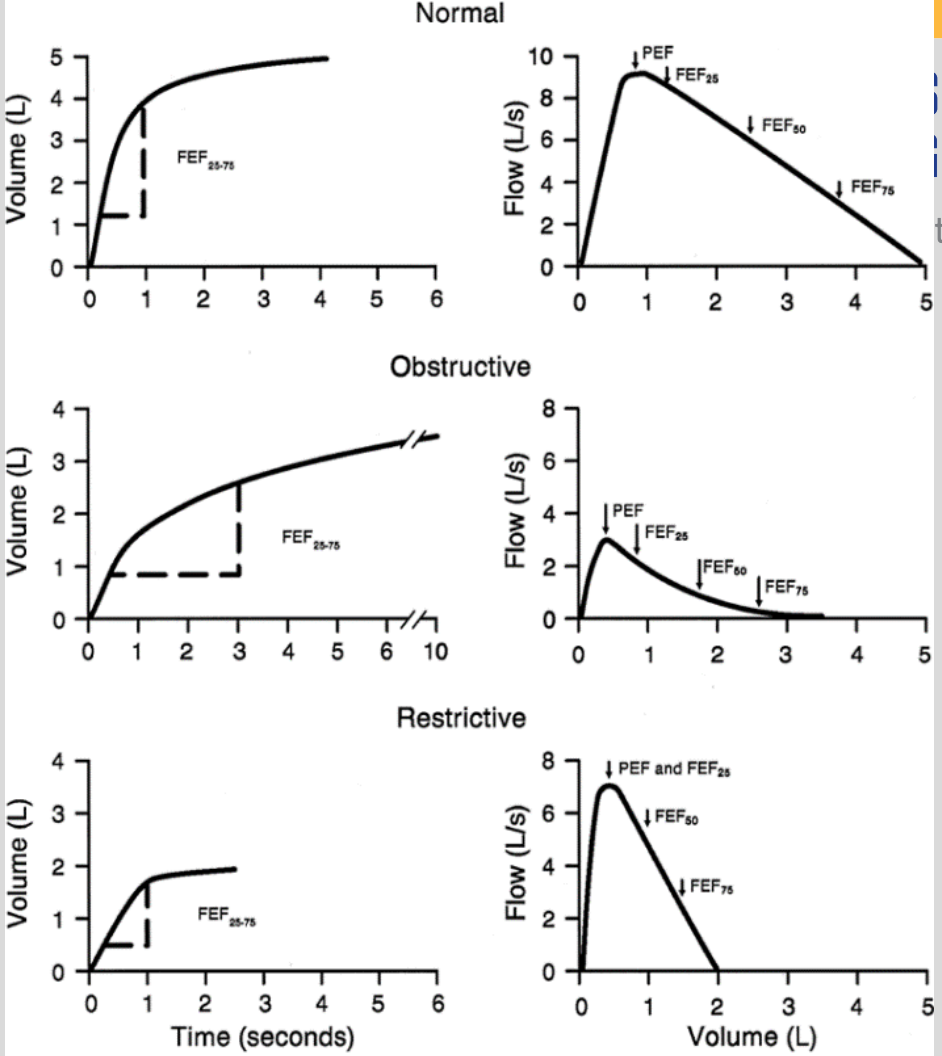
College of Osteopathic Medicine



pathic


Airflow
Inspiration Expiration
TLC
RV
Volume
Fixed obstruction
(intra- or extrathoracic)
A
TLC
RV
Variable intrathoracic
obstruction
B
TLC
RV
Variable extrathoracic
obstruction
C

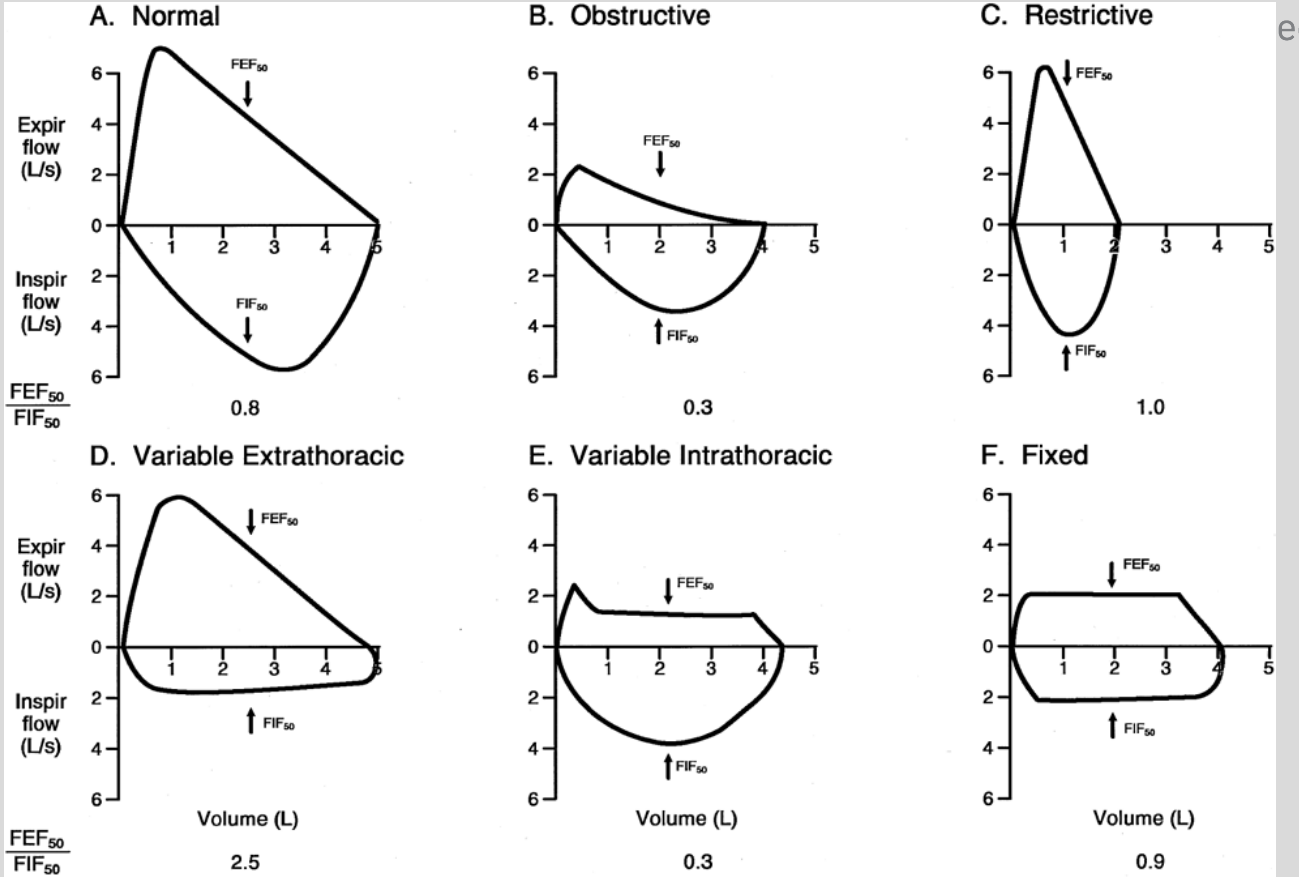
# Reference


Normal
Volume (L)
FEF25-75
Flow (L/s)
PEF
FEF25
FEF50
FEF75
Obstructive
Volume (L)
FEF25-75
Flow (L/s)
PEF
FEF25
FEF50
FEF75
Restrictive
Volume (L)
FEF25-75
Time (seconds)
Flow (L/s)
PEF and FEF25
FEF50
FEF75
Volume (L)

# More Reference

INSTITUTE
LOGY

eopathic



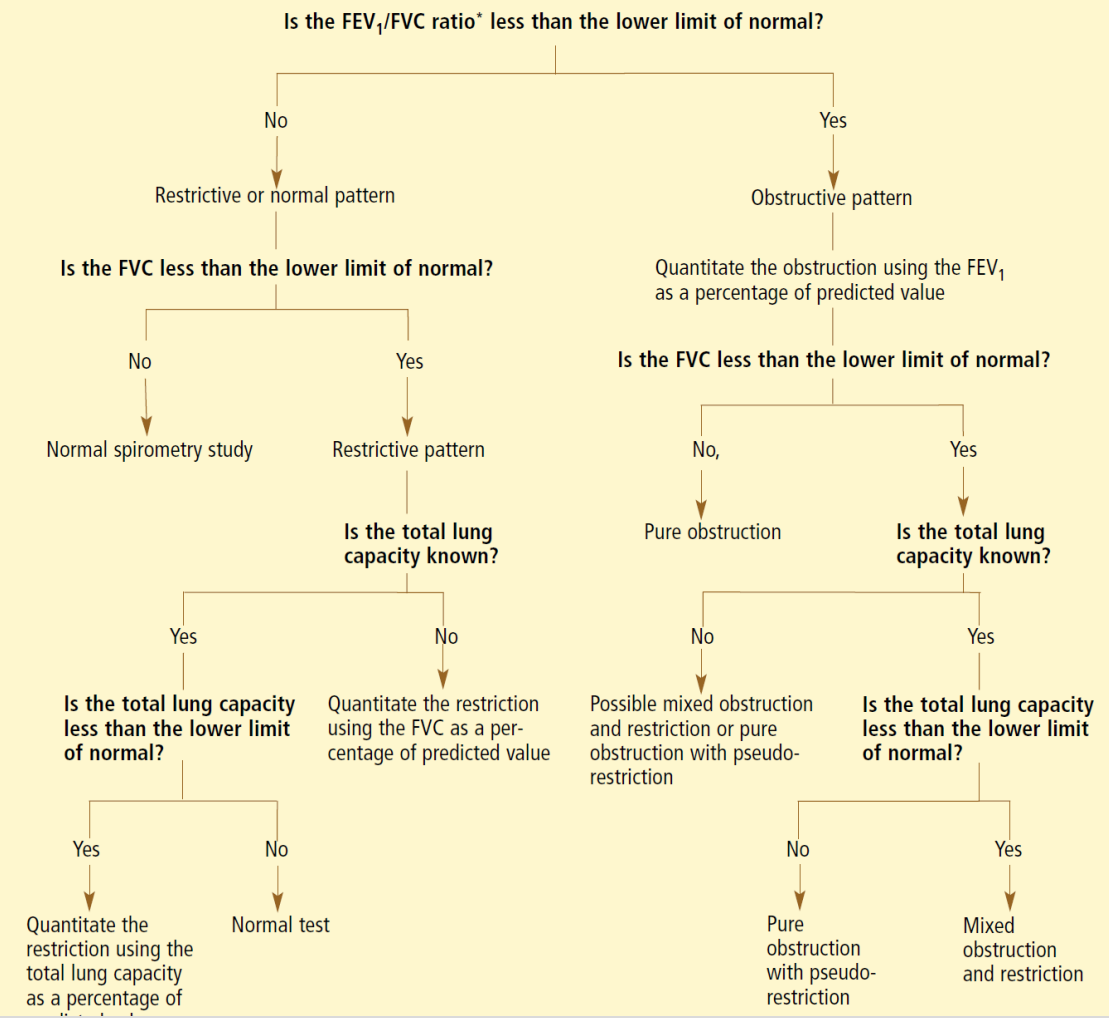
# More Reference

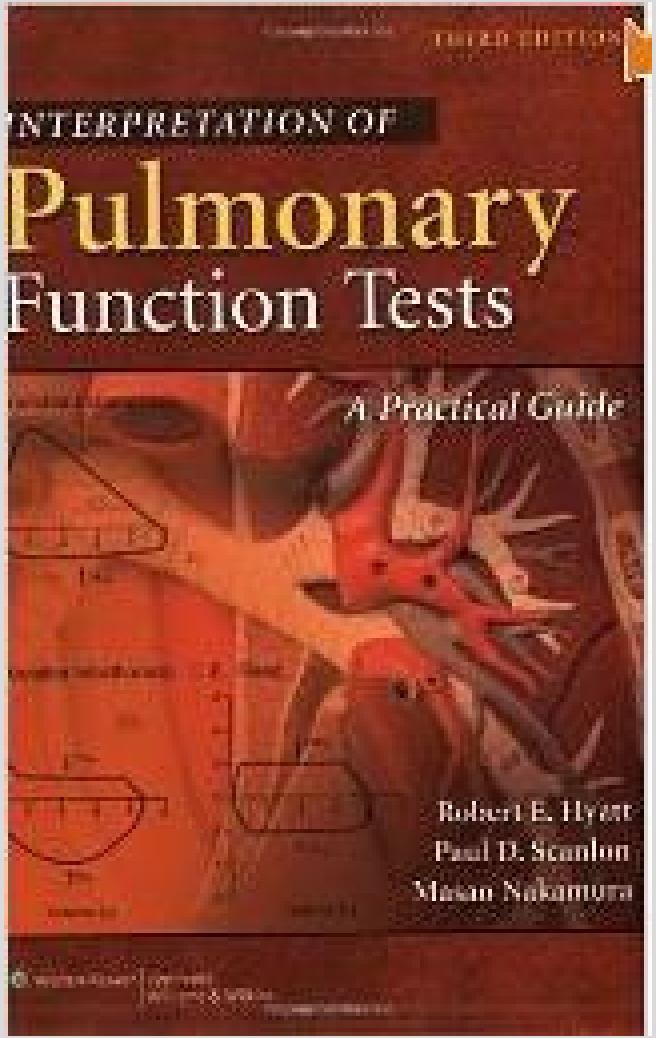
College of Osteopathic Medicine

## Obstructive and restrictive patterns

| MEASUREMENT | OBSTRUCTIVE PATTERN | RESTRICTIVE PATTERN |
|---|---|---|
| **Forced vital capacity** (FVC) | Decreased or normal | Decreased |
| **Forced expiratory volume in 1 second** ($FEV_1$) | Decreased | Decreased or normal |
| **$FEV_1$/FVC ratio** | Decreased | Normal |
| **Total lung capacity** (TLC) | Normal or increased | Decreased |

YORK INSTITUTE
CHNOLOGY
e of Osteopathic
ne






NEW YORK INSTITUTE

Y

thic



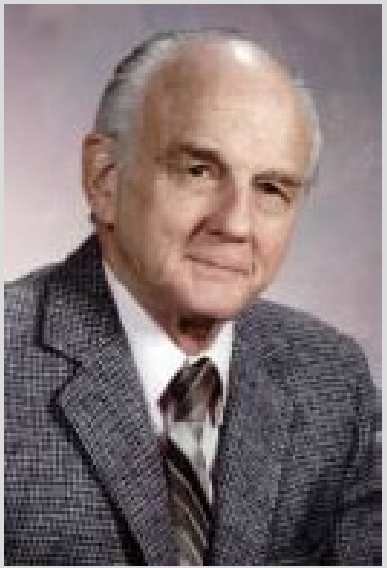
*Journal of Clinical Investigation*
*Vol. 42, No. 1, 1963*

# THE PRESSURE–FLOW RELATIONSHIPS OF THE INTRATHORACIC AIRWAY IN MAN *

By ROBERT E. HYATT † and ROGER E. WILCOX

(*From the Cardiopulmonary Laboratory, Department of Medicine and Department of Surgery, Beckley Memorial Hospital [Miners Memorial Hospital Association], Beckley, W. Va.*)

(Submitted for publication July 6, 1962; accepted September 1, 1962)



RK INSTITUTE
INOLOGY
Osteopathic

NEW YORK INSTITUTE
OF TECHNOLOGY
steopathic

**“I believe we are traveling the road to elucidation of all disease, but the journey’s end may be a long way off.”**

*Charlie Mayo, MD*

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

# THANK YOU

# Lecture Feedback Form:

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R